



Outsourcing and Public Expenditure: Evidence from Contract-Level Data in European Public Procurement (2011–2021)

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Abstract

Public sector outsourcing, promoted under New Public Management reforms from the late 1970s as a means to increase efficiency and reduce public spending, has expanded in the past two decades from technical support functions to a broader range of core public services. Despite its diffusion, evidence on its fiscal implications remains inconclusive, with findings varying across studies and levels of analysis. This thesis examines whether within-country changes in outsourcing intensity are associated with changes in primary government expenditure in the European Union.

The analysis uses micro-data from the Global Public Procurement Database (GPPD), aggregated into a harmonised panel of 27 EU member states from 2011 to 2021. A micro-founded indicator of outsourcing intensity is constructed from awarded contract values and linked to Eurostat fiscal data, institutional quality measures, and political and macroeconomic controls. A two-way fixed-effects model is employed to identify within-country variation while accounting for unobserved heterogeneity and common shocks.

Results show a modest but robust positive elasticity between outsourcing intensity and real primary expenditure. Additional analyses highlight associations with procurement composition, particularly subcontracting and foreign supplier involvement, while economic and functional decompositions reveal no clear sectoral channels. The findings provide contract-level evidence on the association between outsourcing and public spending during a period marked by the post-2008 financial crisis, subsequent austerity measures, and the fiscal pressures of the COVID-19 pandemic.

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Chapter 1

Introduction

Public sector outsourcing, defined as the delegation of public activities to external providers (OECD), has become a structural component of state organisation in many advanced economies. Its emergence is closely linked to the trajectory of New Public Management (NPM), whose reforms diffused widely from the 1980s onwards and reshaped public sectors across sectors and levels of government. These reforms constituted a global policy paradigm, often supported by broad cross-party consensus, and gradually institutionalised contractual logics within public service production (Weghmann and Sankey, 2022). Within this trajectory, outsourcing expanded from a managerial instrument for ancillary services into a mechanism for reorganising state capacity.

Over the past two decades, this evolution has accelerated and diversified. Outsourcing has extended from low-value and technical functions to complex welfare domains such as healthcare, social services and related high-intensity services. Institutional analyses show a shift toward multi-layered provider networks and consultancy-driven contracting, extending private involvement beyond operational tasks into policy, IT and HR functions (Weghmann and Sankey, 2022). These developments reflect a broader process of organisational disembedding, in which the state withdraws from the knowledge, coordination and learning processes underlying public value creation.

The implications of this transformation for public finances are theoretically and empirically ambiguous. Although outsourcing was historically promoted as a means to reduce expenditure, the literature identifies multiple and conflicting mechanisms that prevent any clear *ex ante* fiscal prediction. Empirical studies mirror this complexity, typically finding modest and heterogeneous associations at aggregate levels. In a European context marked by diverse administrative traditions and varying reliance on external providers, assessing how changes in outsourcing intensity relate to primary public expenditure remains a timely and analytically salient question.

The present analysis is situated within the macro-aggregate empirical literature examining the relationship between public sector outsourcing and public expenditure. It contributes to this strand of literature by analysing the directional association between within-country changes in outsourcing intensity and changes in primary public expenditure across the 27 EU member states from 2011 to 2021.

The empirical approach relies on the recently released Global Public Procurement Database (GPPD), which compiles information on more than 72 million public procurement contracts (Fazekas and Tóth, 2021). The granularity of these microdata is a central advantage for the present analysis, as it allows for the construction of harmonised, micro-founded country-year indicators of outsourcing intensity and for the inclusion of procurement characteristics that capture structural differences in procurement practices. The dependent variable is the demeaned logarithm of real primary public expenditure. The main explanatory variable is a country-year indicator of outsourcing intensity derived from the GPPD. For each country and year, outsourcing intensity is computed as the logarithm of the mean value of awarded contracts. To focus identification on within-country dynamics, the indicator is demeaned by its country-specific long-run average. The specification controls for macroeconomic conditions, institutional quality, and government ideology.

A two-way fixed-effects specification is employed to isolate within-country variation while controlling for unobserved time-invariant heterogeneity and shocks common to all member states.

The empirical analysis identifies a modest but statistically significant positive elasticity between within-country changes in outsourcing intensity and real primary public expenditure. This association proves stable across specifications. Robustness checks incorporating procurement composition variables reveal additional systematic patterns: higher subcontracting intensity and a greater incidence of EU and non-EU suppliers are associated with higher expenditure, while years in which procurement shifts toward service rather than goods contracts exhibit a marginally lower level of primary spending. Decomposition exercises using ESA economic categories and COFOG functions do not reveal statistically significant sectoral patterns.

The thesis is structured as follows. Chapter 2 situates the fiscal question within the institutional, theoretical and empirical literature on public sector outsourcing. Chapter 3 introduces the data sources, describes the construction of the main variables, and presents descriptive statistics. Chapter 4 sets out the econometric framework and motivates the identification strategy. Chapter 5 reports the empirical results, including robustness exercises and decomposition analyses, and develops their interpretation in light of theory and prior evidence. Chapter 6 concludes.

Chapter 2

Relevance of the Topic

The process of outsourcing in the public sector has consistently increased over the last decades, deeply transforming the structure and characteristics of public work and employment. This trend has been recently confirmed by the *OECD Government at a Glance 2025* report, which shows that, across OECD countries, outsourced expenditure amounted to 9.2% of GDP in 2023 — a figure that rose by about 0.1 percentage points in the following year (2024). This provides further evidence of the structural expansion of externalised service provision (OECD, 2025).

Understanding the current shape of outsourcing and its impact on public expenditure requires taking into account several interrelated dimensions. The purpose of this chapter is to explore the evolution and relevance of public sector outsourcing as a political–economic issue. It first traces how outsourcing has developed over the past decades (Section I), and then examines economic efficiency as the main rationale for introducing outsourcing, highlighting the mixed results emerging from empirical assessments (Section II).

The discussion of these dimensions situates the topic within its broader historical and political context, as part of the transformation of contemporary capitalism. It also links to the ongoing empirical debate on public sector efficiency. Taken together, these aspects underline the importance of the theme from multiple perspectives and lead to the final discussion of the research question and guiding hypothesis of the study (Section III).

2.1 The Origins of Public Sector Outsourcing

The expansion of outsourcing in Western public sectors is rooted in the broader political–economic environment defined as *neoliberalism*. Traditionally conceived as a phenomenon supporting the transfer of state functions to market actors, this environment has driven a structural reconfiguration of public service provision. A leading assumption within this con-

text is that public administrations are inherently inefficient, oversized, or resistant to change. This presumption has legitimised repeated calls for trimming, restructuring, privatising, or contracting out public functions, promoting organisational models that resemble private businesses (Weghmann and Sankey, 2022).

A first important phase in this trajectory was the wave of large-scale privatisation programmes implemented across many Western countries during the 1980s and 1990s. These reforms redefined the boundaries between state and market and opened significant portions of public service provision to private actors. As widely recognised in the public services literature, privatisation and marketisation evolved together, with the latter generating the economic and ideological conditions that supported an expansion of private involvement in public functions (Weghmann and Sankey, 2022).

Within this process, outsourcing emerged as a subsequent phase: not merely an administrative technique, but a form of functional privatisation through which state activities are transferred to external providers. In the decades following the Reagan and Thatcher administrations, this shift accelerated markedly, contributing to the structural growth of contracting-out across Western public sectors (Collington and Mazzucato, 2024).

It is against this background that New Public Management (NPM) gained prominence as a managerial reform agenda, particularly in Anglo-Saxon countries. From the 1990s onwards, the NPM literature promoted a set of organisational reforms intended to enhance efficiency, accountability, and measurable results. These reforms included greater budget transparency, the replacement of hierarchical structures with contractual relationships, the disaggregation of large agencies into smaller units, the creation of quasi-markets (such as purchaser-provider splits), and the introduction of competition mechanisms (Dunleavy and Hood, 1994). Through these principles, market logic was further embedded in the public sector, reinforcing the trajectories opened by the earlier wave of privatisation. This translated into concrete reforms in the provision of frontline goods and services, as well as increased reliance on private-sector expertise in key administrative roles (Collington and Mazzucato, 2024).

From the early 2000s onwards, a new phase of public management reform emerged, commonly referred to as post-NPM. While several core aspects of NPM—such as marketisation and managerial tools—were preserved, additional elements were introduced, including a renewed emphasis on impartiality, associated with Neo-Weberian approaches, and the integration of digital governance tools. Rather than replacing NPM, the post-NPM phase complemented it by adding greater coordination, centralisation, and collaborative structures (Lapuente and de Walle, 2020).

Following the economic and financial crisis of 2008, the political agenda in many Western

countries again prioritised the reduction of the public sector’s size. The objective was no longer primarily to reform public administrations along managerial lines, but to curtail public expenditure and achieve fiscal consolidation (Alonso et al., 2011). In this context, fiscal pressure became a major driver of market-oriented reorganisation (Pollitt and Bouckaert, 2017).

These transformations were not merely technical or bureaucratic adjustments but reflected a deeper redefinition of the role of the state that had already begun during the privatisation wave of the 1980s and 1990s. The progressive minimisation and restructuring of public institutions in that period laid the groundwork for a longer-term shift towards the marketisation of public functions. The post-2008 austerity cycle intensified this trajectory, reinforcing the pursuit of fiscal restraint as a guiding principle of state reform (Alonso et al., 2011).

This shift forms part of a broader and distinct phase in the evolution of contemporary capitalism, commonly described as *rentified capitalism*. In this phase, profit accumulation no longer depends primarily on production but increasingly on the control of access, the redefinition of property rights, and the monetisation of previously public or common goods (Dosi et al., 2024). Within this framework, market logic has progressively extended to domains that were once considered collective or universal, such as health, education, and welfare — through a process known as *commodification*.

In the process of *commodification*, public goods and services are treated as commodities and converted into private rent-bearing assets. This transformation, however, is far from straightforward: it requires redefining such goods as divisible and excludable, so that they can be exchanged and priced within market frameworks. The subsequent process, whereby distinctions between goods are removed to enable economic competition, is referred to as *commoditization* (Spicker, 2024).

Ultimately, once commodified, these goods and services are introduced into market relations — a process known as *marketization*. According to Spicker (2024), this can occur in three main forms: first, through the *sub-contracting* of public service activities to private firms, under the assumption that they operate more efficiently; second, by *marketizing* public services internally, introducing mechanisms such as competition and performance incentives; and third, by developing a *mixed economy*, in which marketised and residual public services coexist. The latter configuration, in particular, has become the dominant one in contemporary welfare systems, where governments increasingly act as residual providers of last resort.

In summary, the evolution of public sector outsourcing cannot be understood as a neutral administrative adjustment. It represents a structural shift in the political economy of the state, where fiscal discipline, market logic, and managerial practices converge. From the

neoliberal reforms of the 1980s to the austerity-driven transformations following the 2008 crisis, outsourcing has been progressively embedded in a broader process of commodification of public functions. Within this framework, the state increasingly acts as a regulator and purchaser rather than as a direct provider of services, while the boundary between public value and private profit becomes increasingly blurred.

2.2 The Fiscal Logic of Outsourcing: Expected Mechanisms and Empirical Evidence

In the previous section, we have briefly discussed the theoretical background and the evolution of public sector outsourcing.

In this way, we have underlined how the diffusion of contracting out has been the result of a broader transformation, involving economic theory, political change, and managerial innovations, through which the State has been progressively emptied of its traditional role. This is quite emblematic if we consider that in the 1960s the Great Society was conceiving governments as all-purpose direct service providers, whereas just a couple of decades later, under the Reagan administration, public services were increasingly relying on external, private actors, in the direction of what has been described as the "hollow state" (Brown and Potoski, 2003).

This transformation can be treated as a first, broad conceptual rationale behind outsourcing practices: conceiving the State and public administration as unable to fulfil their mission — in a certain sense, infantilizing them (Mazzucato and Collington, 2023) — serves to legitimise the delegation of public functions to private actors.

Importantly, this trajectory has not unfolded uniformly across advanced economies. Outsourcing expanded earliest and most extensively in Anglo-Saxon liberal market economies, whereas continental coordinated market economies followed more gradual and selective paths. Differences in administrative traditions, state–labour relations, and institutional complementarities shaped the extent and form of externalisation, producing nationally specific configurations of public–private arrangements (Weghmann and Sankey, 2022).

Given the relevance of the topic, a substantial body of research has investigated the purposes that lead governments to outsource public functions, and most of the available evidence explains the phenomenon as a combination of political, economic, and managerial rationales.

To align with the purpose of this paper, here we focus on two approaches: the first adopts the lens of economic theory, emphasizing how *ex ante* competition may reduce costs

and increase productivity in outsourced services; the second follows the literature on public policy and the role of the State, addressing issues such as risk management and accountability in contracting-out practices (Domberger and Jensen, 1997).

The stream of literature following the economic approach justifies contracting out as a means to reduce public sector costs through competitive efficiencies and economies of scale (Brown et al., 2010). Indeed, differently from privatization, the process of outsourcing involves a series of features which are expected to be responsible for cost-cutting and performance improvement.

First of all, contracting out implies introducing a mechanism of *ex ante* competition, as opposed to competition in the market: different private firms compete in a market constituted by the contract specification, and the winner of the auction process is the firm offering the lowest prices (or the Most Economically Advantageous Tender). (Domberger and Jensen, 1997)

In this sense, the first mechanism through which contracting out is expected to reduce public costs is by enabling firms to compete within a regulated setting: such competition should, in principle, lead to cost reductions through the selection of the most efficient providers of the required goods or services. Similarly, managerial literature often underlines how private employees operate with profit-oriented incentives, which seem more appropriate to pursue innovation and efficiency even within the public sector (Brown et al., 2010).

From a theoretical standpoint, many limitations to this precise justification have been highlighted. For example, still within the mainstream perspective on competition, it is often argued that private firms and nonprofits avoid entering competition for public sector contracts because of the complexity and time-consuming nature of legal procedures: this significantly reduces the number of competing firms, and therefore the potential efficiency of the selection process (Brown et al., 2010).

Additionally, differently from private firms, governments tend to have “ambiguous” goals and significant constraints: public administrations pursue broader objectives than cost savings or profit generation. Other aims — such as equity — may be pursued due to political or ideological reasons, and these values may be conflicting and cannot necessarily be addressed through segmentation or the weakening of public structures (Brown et al., 2010).

Finally, when discussing the economic efficiency of contracting out, orthodox literature focuses extensively on transaction costs (see (Williamson, 1975), (Williamson, 1985)). This theory highlights how outsourcing practices involve a range of costs due to the legal and institutional frameworks in which they are embedded: drafting contract specifications, evaluating tenders, or negotiating the final agreement with the selected provider all entail administrative burdens (Domberger and Jensen, 1997). According to this approach, contracting out is

efficient only if it is capable of offsetting these administrative and legal costs.

A second stream of literature, grounded in public policy analysis, explains public sector outsourcing in terms of political accountability and risk management (Brown and Potoski, 2003).

Within this framework, accountability emerges as a central theme in justifying the delegation of public service provision to private actors. One of the main advantages often associated with contracting out is that public authorities retain formal control over outsourced activities through contractual monitoring tools, enabling them to supervise performance and ensure compliance (Domberger and Jensen, 1997).

From this perspective, shifting operational accountability from public agencies to private contractors is seen as beneficial, as it allows governments to exercise greater control over both costs and service quality (Blomqvist and Winblad, 2020). Nonetheless, the literature has identified several critical limitations to this rationale. First, accountability in the public sector differs fundamentally from accountability in market settings. Monitoring the performance of private contractors in public services is not merely a technical matter but also a question of democratic legitimacy (Blomqvist and Winblad, 2020). Accountability, in this context, involves the ability to hold actors responsible for their actions — a core value in democratic governance.

Blomqvist and Winblad (Blomqvist and Winblad, 2020) distinguish at least three forms of accountability. The first is hierarchical accountability, typical of public organizations, where direct control is exercised through legal or regulatory mechanisms. The second is market-based accountability, which relies on price signals and competition as proxies for performance. The third is social accountability, which rests on mutual relationships and reciprocal expectations between actors.

The introduction of market-based accountability into the public sphere can generate significant tensions. Specifically, it may weaken the capacity of elected officials to be held directly accountable for the quality and equity of public service delivery. As a result, the relationship between political representatives and service provision becomes more opaque, diminishing democratic responsiveness (Blomqvist and Winblad, 2020).

Within the literature on public sector outsourcing, risk management has been increasingly recognised as a central rationale underpinning contracting-out decisions, especially in complex service delivery. As (Bloomfield, 2019) notes, risk allocation is not merely a legal or technical issue but is embedded in the design of public contracts through standardised terms and conditions aimed at mitigating recurrent vulnerabilities. In this sense, outsourcing is understood as a tool for reallocating operational and financial risks from the public sector to private actors. This process, however, requires careful *ex ante* planning and contract design

to ensure that responsibilities and liabilities are clearly assigned and managed throughout the life of the contract.

While theoretical accounts provide a solid foundation for understanding the rationale behind public sector outsourcing, emphasising mechanisms such as competitive pressure, efficiency gains, and enhanced accountability, important questions remain regarding the actual fiscal outcomes of such practices. In particular, the extent to which outsourcing leads to measurable reductions in public expenditure remains subject to debate.

Governments have frequently justified outsourcing as a tool for containing or reducing costs, particularly in the context of fiscal consolidation strategies pursued in the aftermath of economic crises. This rationale has been endorsed by international organisations such as the OECD and the European Commission, which have actively promoted the adoption of outsourcing policies across member states (Alonso et al., 2017). Nevertheless, empirical assessments of outsourcing’s fiscal impact have yielded mixed and, at times, inconclusive results.

While a number of studies report positive effects in terms of cost control and efficiency, particularly when *ex ante* competition mechanisms are properly implemented, other contributions point to several countervailing dynamics. These include transaction and monitoring costs, the influence of interest groups, and the challenges of maintaining public accountability and service quality (Hodge, 2000; Bel and Warner, 2015; Brown and Potoski, 2005; Hefetz and Warner, 2012).

In light of this debate, here we aim to review the empirical literature on outsourcing and public spending, with the purpose of assessing whether, and under what conditions, outsourcing can effectively contribute to the reduction of public expenditure.

The empirical literature on public sector outsourcing is extensive yet fragmented. Scholars from various academic backgrounds have acknowledged the relevance of this topic and have explored it through multiple empirical lenses, ranging from aggregate analyses to sectoral and regional investigations. Notably, there is a lack of consensus regarding the empirical impact of outsourcing on public expenditure. Moreover, the results often depend on the level of analysis, institutional settings, and sectoral context.

At the macro level, empirical studies examining the relationship between outsourcing and aggregate public expenditure reveal mixed and often ambiguous results. As discussed in the previous section, advocates of public sector outsourcing argue that contracting out services reduces public expenditure by introducing competitive pressures and the efficiency incentives of private ownership. These are expected to result in cost savings and increased productivity. Critics, however, highlight potential offsetting effects, including high transaction costs, moral hazard, fiscal illusion, and deterioration in service quality or employee working condi-

tions—commonly referred to as the "quality shading" or "redistribution" hypotheses.

Early studies on public sector outsourcing predicted reductions in public expenditure ranging from 15–20 per cent, with some estimates as high as 30 per cent. However, more recent aggregate-level analyses have significantly revised these projections downward (Petersen et al., 2018). A systematic series of international studies conducted between 2000 and 2014 found that average cost reductions during this period amounted to just 4.2 per cent. Furthermore, these savings declined over time—from 8.5 per cent during 2000–2004 to a mere 0.4 per cent over the full 2000–2014 period (Petersen et al., 2018). These findings reflect a more tempered assessment of outsourcing’s cost-efficiency, based on robust cross-country empirical evidence.

At the central government level, several aggregate studies do not confirm the anticipated cost reductions. For example, Alonso et al. (2017) analyzed 25 EU countries between 1990 and 2011 and found that outsourcing was not associated with a reduction in public expenditure. On the contrary, their findings indicate a positive and statistically significant relationship between outsourcing and government spending, with coefficients ranging from 0.13 to 0.21. The authors conclude that while outsourcing may yield savings in specific sectors, these do not necessarily translate into aggregate expenditure reductions at the central level. Hence, the claim that outsourcing is inherently cost-efficient cannot be generalized (Alonso et al., 2017).

In the same vein, a study examining European countries over the 1980–2010 period identified a short-term positive correlation between outsourcing and public expenditure, aligning with the theory of transaction costs (Alonso et al., 2011). Over the long term, the effect was either positive or statistically insignificant, consistent with the "hold-up" hypothesis, which posits that initial savings may not be sustained.

Sectoral-level analyses reveal divergent outcomes, indicating that outsourcing’s effectiveness depends heavily on transaction costs related to the contracting process (Petersen et al., 2018). For instance, at the regional level in Spain, outsourcing was associated with increased consumption and healthcare expenditure, while it had no significant impact on investment or education spending (Delgado-Téllez et al., 2022).

The mainstream literature often distinguishes between technical services (which are relatively easy to contract) and soft services (which present greater challenges in terms of contractualization). Technical services—such as waste collection, cleaning, road maintenance, and water services—tend to exhibit more consistent cost-saving outcomes. For instance, empirical studies on waste collection report average savings of 22 per cent without compromising service quality (Jensen and Stonecash, 2005; Andersson et al., 2019). Likewise, a U.S.-based analysis of the dredging sector reported a 23 per cent cost reduction due to out-

sourcing (?). A study conducted in Spain found that outsourced social services (via indirect provision) were more cost-effective than publicly managed alternatives (Chow et al., 2025).

Conversely, outcomes in social and welfare services, which involve higher complexity, labor intensity, and transaction costs, tend to be less favorable. A systematic review covering the 2000–2014 period found no conclusive evidence of cost-efficiency in outsourcing elderly care, healthcare, employment services, or prison management (Petersen et al., 2018). In these areas, empirical support for outsourcing’s economic advantages is limited, though isolated studies report savings in sectors like employment placement and prison management (Chow et al., 2025).

Heterogeneous results also emerge from analyses at the regional or local levels. In these contexts, institutional and geographical variations play a crucial role. For example, a review of empirical studies from 2000–2014 identified systematic differences between Anglo-Saxon and continental European countries: documented cost savings in Anglo-Saxon countries averaged 6.4 per cent, more than double the 2.9 per cent observed elsewhere (Petersen et al., 2018). These disparities are attributed to institutional and regulatory differences, such as more liberal labor markets and weaker trade unions in Anglo-Saxon countries, which may amplify outsourcing’s effects (Petersen et al., 2018). A more recent regional-level study in Spain (2002–2018) found a positive correlation between public sector outsourcing and public expenditure, echoing macro-level findings. This study reported an increase in public spending of 2.9 per cent in the short term and 5.5 per cent over the long term. The impact was particularly pronounced during periods of fiscal consolidation (post-2007), suggesting that austerity may have influenced the outsourcing process. Notably, Catalonia was the only region where a negative relationship was observed, potentially reflecting the role of economies of scale in regions with high levels of outsourcing (Delgado-Téllez et al., 2022).

Overall, the literature offers a fragmented and inconclusive account of the fiscal implications of public sector outsourcing. While theoretical frameworks point to potential efficiency gains and cost savings, empirical findings are mixed and strongly shaped by institutional, sectoral, and administrative conditions. Observable savings tend to emerge in technical services with low transaction costs, whereas complex or relational services often deliver uncertain or even adverse fiscal effects.

At the macro level, cross-country studies rarely confirm consistent reductions in public expenditure, frequently highlighting countervailing factors such as transaction costs, limited competition, and weakened public accountability. Despite a substantial body of research and the centrality of outsourcing in reform agendas promoted by international organisations, the overall evidence remains ambiguous and contested. This calls for further investigation through updated, comparative, and context-sensitive analyses — especially in view of the

fiscal and institutional transformations that have characterised the post-crisis decade.

2.3 Questioning the Fiscal Promise: Framing the Research Hypothesis

In the previous sections, we have situated public sector outsourcing at the intersection of theoretical, institutional, and empirical trajectories. We have highlighted its centrality as a marker of structural transformations in the role of the state and analysed the economic and managerial rationales that have supported its diffusion in recent decades. These insights have guided our review of the empirical literature on the relationship between outsourcing and public expenditure, with the goal of answering two interrelated questions: (i) does contracting out effectively reduce public spending? and (ii) under which conditions—institutional, political, sectoral, or contractual—does this relationship vary?

Although a substantial body of research has addressed these issues, findings remain inconclusive and context-dependent. As reviewed, consistent fiscal savings are more likely in technical services with low transaction costs, while more complex sectors often show negligible or adverse fiscal effects. Moreover, aggregate-level studies rarely confirm the expected cost reductions, pointing instead to offsetting mechanisms, such as limited competition, institutional weaknesses, or governance failures. This ambiguity is particularly salient in the context of the post-crisis decade, where outsourcing has been promoted as part of broader fiscal consolidation and public management reforms across Europe.

This research contributes to this ongoing debate by providing a cross-national econometric analysis of the link between outsourcing intensity and public expenditure across the 27 EU countries over the period 2011–2021. Rather than testing a single directional hypothesis, the objective is to evaluate whether—and under what conditions—outsourcing is systematically associated with changes in public expenditure levels and composition.

A baseline fixed-effects model is estimated using a novel macro dataset derived from the Global Public Procurement Database (GPPD), in combination with Eurostat fiscal indicators and institutional/political controls. The main explanatory variable is a country–year measure of outsourcing intensity, constructed from micro-level contract data. The baseline dependent variable is real primary public expenditure (log-transformed and demeaned), while robustness analyses consider alternative outcome variables based on the economic (e.g. intermediate consumption, personnel costs) and functional (e.g. health, education) composition of spending.

In addition, the analysis accounts for procurement-specific features—such as contract

type, award criteria, subcontracting, and supplier origin—which may mediate the outsourcing–spending relationship. These variables are included as compositional controls to isolate structural differences in procurement practices and test whether the estimated effects are robust to procurement heterogeneity.

By integrating detailed procurement data with harmonised fiscal statistics and governance indicators, this study aims to identify systematic patterns in the directional association between outsourcing intensity and public spending. In doing so, it provides a rigorous, data-driven contribution to the empirical literature and a basis for more advanced modelling strategies. It should therefore be regarded as a preliminary yet methodologically grounded examination of a central question in contemporary fiscal governance.

Chapter 3

Data and Descriptive Statistics

This chapter presents the data employed in the empirical analysis, describing the sources, the construction of the main indicators, and the summary statistics of the main variables. The objective is to provide the reader with a comprehensive overview of the dataset and to offer preliminary insights into outsourcing and public expenditure patterns across European countries between 2011 and 2021.

3.1 Data Sources

The empirical dataset is an unbalanced country–year panel covering the 27 EU member states over the period 2011–2021. The unit of analysis is the country–year, identified by the two-letter Eurostat country code. After data harmonisation and cleaning, the dataset includes 231 usable annual observations.

The analysis combines information from multiple sources to ensure temporal coherence and analytical consistency. Contract-level data are drawn from the *Global Public Procurement Database* (GPPD) to build indicators of outsourcing intensity and dynamics; data on public expenditure come from Eurostat; the World Bank’s *Worldwide Governance Indicators* (WGI) provide measures of institutional quality; and the *Manifesto Project Database* supplies information to construct the political orientation indicator.

The recently released GPPD (Fazekas and Tóth, 2021) compiles information on more than 72 million public procurement contracts across 42 countries from 2006 to 2021. It provides highly granular microdata on individual public contracts, including detailed information on the buyer(s) and supplier(s), geolocation, product classification (CPV codes), contract value, and key elements of the procurement process, such as the award date, procedure type, and contracting authority.

This level of detail represents a major analytical advantage: it allows the construction

of macro-level indicators from micro-level transactions, making it possible to disentangle structural patterns in public sector outsourcing. In particular, the dataset enables the measurement of outsourcing intensity by aggregating procurement expenditures over time and across sectors, providing an innovative and data-rich foundation for analysing the relationship between outsourcing and public expenditure.

To ensure cross-country comparability, all monetary values in the GPPD are converted to a common base using Purchasing Power Parity (PPP) adjustments. This correction accounts for price-level differences across countries and over time, thereby allowing consistent comparison of public spending magnitudes. Additionally, microdata were filtered to include only finalised tenders and winning bids (`tender_isawarded = TRUE`, `bid_iswinning = TRUE`), avoiding duplication and non-final awards.

After these filters were applied, the working dataset contained **2.3 million contracts** across EU-27 countries.¹

To ensure sufficient coverage for each country, the dataset is restricted to the period 2011–2021 and to EU-27 member states. This temporal scope was selected to ensure a more balanced panel and consistent data coverage across countries. Earlier years were excluded due to substantial asymmetries in data completeness and reporting standards, which would have introduced structural bias into longitudinal comparisons. As shown in Appendix A.1, data coverage becomes stable and comprehensive from 2011 onwards, both in the number of contracts and in total procurement value. The restriction to EU-27 countries also ensures greater homogeneity in institutional characteristics, public spending patterns, and legal frameworks.

After data cleaning, individual contract records were aggregated at the *country-year* level, generating a macro dataset of procurement-based indicators suitable for econometric analysis.

Macroeconomic and fiscal data are sourced from *Eurostat*, the statistical office of the European Union and the main repository for harmonised government finance statistics across member states. The Eurostat database provides consistent time series on public expenditure and its main components, compiled under the European System of Accounts (ESA 2010).

These data constitute the backbone of the macro-fiscal dimension of the analysis, supplying the measures of total and primary public expenditure. Eurostat’s comprehensive coverage enables the decomposition of government spending both by economic category and by functional purpose, according to the international COFOG classification. This dual disaggregation makes it possible to analyse the structure of public expenditure alongside its

¹The precise number of retained contracts and their distribution by country and year are reported in Appendix A.1.

evolution over time.

In addition, Eurostat provides the deflators and national accounts aggregates used to express all fiscal variables in real terms and to normalise them relative to GDP. This ensures temporal and cross-country comparability, aligning the macroeconomic data with the PPP-adjusted procurement figures drawn from the GPPD. Overall, Eurostat’s harmonised methodology and data consistency make it the most reliable source for cross-national analysis of public finances within the European Union.

Institutional quality indicators are taken from the *World Bank’s Worldwide Governance Indicators* (WGI), which provide internationally comparable measures of governance performance. Among the six available dimensions, the *control of corruption* index is employed as a proxy for the quality of public institutions. This choice is consistent with the empirical literature on public sector efficiency and governance, where this indicator is widely adopted as a summary measure of administrative integrity and institutional effectiveness across countries. It captures the extent to which public power is exercised for private gain and thus reflects one of the most relevant institutional dimensions affecting procurement outcomes and fiscal performance.

Political data are obtained from the *Manifesto Project Database* (MPD), developed and maintained by the Manifesto Research on Political Representation (MARPOR) group at the Social Science Research Center Berlin. The MPD provides harmonised, cross-national information on political parties’ policy positions and electoral outcomes, derived from quantitative content analysis of party manifestos. It is one of the most widely used and recognised datasets in comparative political science.

Table 3.1: Data sources and analytical roles (Part I)

Source			Main variables	Role in analysis
Global Procurement (GPPD)	Public Database	Pro-	<i>tender_digiwhist_price</i> , contract type, award criteria, subcontracting, supplier origin	Micro-level source for outsourcing indicators and procurement composition shares
Eurostat	– Government Finance Statistics (GFS) and National Accounts	Govern-	Total expenditure (TE), interest payments (D.41PAY), nominal GDP, GDP deflator (2010 = 100), real primary expenditure	Fiscal and macroeconomic controls; denominators for normalisation

Table 3.2: Data sources and analytical roles (Part II)

Source	Main variables	Role in analysis
World Bank – World-wide Governance Indicators (WGI)	<i>corruption_control</i> (institutional quality, -2.5 $\rightarrow$ +2.5 scale)	Institutional quality variable used as macro-level governance control
Manifesto Project Database (MP2025a)	<i>gov_state_market</i> (seat-weighted index of state-market orientation of governing parties)	Political orientation indicator capturing government ideology

3.2 Variable Construction

This section provides a detailed description of the variables used in the baseline and robustness analyses, as well as an explanation of how the main indicators are constructed for the empirical investigation. Further information regarding the sources of each variable can be found in Appendix B.

3.2.1 Dependent Variable

The baseline dependent variable is *real primary expenditure*, derived from Eurostat’s Government Finance Statistics (*gov_10a_main*). Primary expenditure is computed as the difference between total expenditure (TE) and interest payments (D.41PAY), thereby excluding debt service costs and capturing the discretionary component of government spending:

$$\text{Primary Expenditure}_{it} = \text{TE}_{it} - \text{D.41PAY}_{it}$$

Nominal values (expressed in millions of national currency) are converted into real terms using the GDP deflator (base year 2010):

$$\text{Real Primary Expenditure}_{it} = \frac{\text{Primary Expenditure}_{it}}{\text{GDP Deflator}_{it}/100}$$

To remove country-level scale effects and focus on within-country variation, the variable is expressed in logarithms and demeaned by its country-specific mean:

$$\tilde{y}_{it} = \ln(\text{Real Primary Expenditure}_{it}) - \overline{\ln(\text{Real Primary Expenditure}_i)}$$

This transformation ensures that the analysis captures temporal changes in fiscal dynamics rather than cross-sectional differences in absolute spending levels, and also mitigates data

skewness.

Empirical studies on fiscal dynamics commonly rely on total public expenditure (Alonso et al., 2017) and expenditure as a share of GDP (Alonso et al., 2011; Delgado-Téllez et al., 2022; Barkley, 2021). Both measures are subject to conceptual and econometric limitations: interest-inclusive expenditure mixes discretionary and debt-service components, while normalising by GDP may induce endogeneity, as public spending itself influences output through fiscal multipliers

3.2.1.1 Decomposition of Public Expenditure

To test the robustness of the results and examine potential heterogeneity in the relationship between outsourcing and spending, several alternative dependent variables are considered. All are obtained from Eurostat, expressed in real terms, log-transformed, and demeaned to maintain consistency with the baseline specification.

(1) Economic Composition of Expenditure. The first robustness group focuses on the economic classification of expenditure by input type (Eurostat, `gov_10a_main`). Three main aggregates are used:

- **P2 – *Intermediate consumption*:** expenditure on goods and services purchased from the market, including outsourced activities.
- **D1 – *Compensation of employees*:** total wage bill of public sector employees, capturing in-house production costs.
- **P5 – *Gross fixed capital formation*:** government investment in infrastructure and other long-term assets.

These indicators allow the testing of whether outsourcing intensity affects specific components of public expenditure differently—for instance, whether higher outsourcing reduces personnel costs (D1) while increasing market purchases (P2). Such decomposition follows the empirical strategy used in recent analyses of fiscal consolidation and expenditure efficiency (Delgado-Téllez et al., 2022).

(2) Functional Composition of Expenditure (COFOG). A second set of alternative variables employs Eurostat’s functional classification (`gov_10a_exp`), based on the international COFOG standard. This breakdown allows for the analysis of whether the relationship between outsourcing and spending varies across policy areas:

- **TF6_1 – General public services,**

- TF6_3 – Public order and safety,
- TF6_4 – Economic affairs,
- TF6_7 – Health,
- TF6_9 – Education,
- TF6_10 – Social protection.

These six COFOG functions are selected because they encompass the most substantial and fiscally relevant components of primary expenditure across EU member states, and represent the policy domains in which outsourcing is structurally most significant.

This approach captures sectoral heterogeneity in fiscal responses, reflecting evidence that the composition of expenditure influences fiscal adjustment dynamics and efficiency outcomes (Delgado-Téllez et al., 2022).

3.2.1.2 Comparability and Scaling

Given that the Global Public Procurement Dataset (GPPD) used to construct the independent variable is adjusted for Purchasing Power Parity (PPP), all expenditure measures are expressed in real terms to ensure cross-country comparability. This adjustment eliminates price-level and size differences, ensuring coherence between micro-level procurement data and macro-level fiscal aggregates.

Overall, the dependent variable and its robustness measures capture multiple dimensions of government spending (aggregate, economic, functional, and decentralised) within a harmonised and comparable empirical framework. All measures are real, log-transformed, and demeaned to ensure consistency with the econometric specification and to focus on within-country temporal variation in fiscal behaviour.

3.2.2 Independent Variable

The explanatory variable is the *intensity of outsourcing*, constructed from the Global Public Procurement Database (GPPD) (Fazekas and Tóth, 2021). This variable captures the relative scale of public contracting activity and serves as a proxy for the degree of market involvement in the delivery of public goods and services.

3.2.2.1 Construction of the Main Indicator

For each country i and year t , outsourcing intensity is measured as the logarithm of the mean value of awarded public contracts:

$$\text{Outsourcing Intensity}_{it} = \ln \left(\frac{\sum_{k=1}^{N_{it}} \text{Contract Value}_{kit}}{N_{it}} \right)$$

where N_{it} is the total number of awarded tenders in that country–year. The use of a logarithmic transformation is motivated by the highly skewed distribution of contract values in the GPPD. Raw procurement amounts display extreme right-tail behaviour, while their logarithmic transformation yields a well-behaved, approximately symmetric distribution. Summary diagnostics are reported in Appendix A. To isolate within-country variation and eliminate cross-country scale differences, the indicator is demeaned by its country-specific average:

$$\tilde{x}_{it} = \ln(\text{Mean Contract Value}_{it}) - \overline{\ln(\text{Mean Contract Value}_i)}$$

This transformation produces a centred variable expressing whether a country in a given year outsourced more or less than its long-term historical average. Positive values indicate years with above-average outsourcing intensity, while negative values signal relative contraction.

3.2.2.2 Price and PPP Adjustments

The GPPD provides all contract values already adjusted for *Purchasing Power Parity* (PPP), expressed in international dollars. This feature enables meaningful cross-country comparisons by neutralising differences in national price levels and exchange rates. As a result, the PPP-adjusted procurement data are directly compatible with the Eurostat fiscal variables expressed in real terms, ensuring consistency.

3.2.2.3 Procurement Composition Controls

The empirical model includes a set of procurement–specific covariates that capture structural heterogeneity in public contracting. These variables summarise, at the country–year level, the composition and characteristics of awarded contracts. These controls are computed as proportions over the total number of awarded and winning tenders and therefore take values in the interval $[0, 1]$. Each indicator represents the share of contracts in a given category, with the omitted group serving as reference. Formally:

$$\text{prop}_{x,it} = \frac{n_{x,it}}{n_{\text{total},it}}$$

where $n_{x,it}$ denotes the number of contracts of type x in country i and year t , and $n_{\text{total},it}$ is the total number of awarded contracts in that country-year.

These compositional controls make it possible to disentangle variation in the intensity of outsourcing from broader changes in procurement structure, ensuring that the estimated effect of outsourcing does not capture shifts in the types of contracts, criteria, or supplier origins.

Table 3.3: Procurement composition variables and reference categories

Category	Variable	Reference Category
Supply type	<code>prop_services</code> , <code>prop_works</code>	Supplies (goods)
Award criterion	<code>prop_meat</code>	Lowest price
Subcontracting	<code>prop_subcontracted</code>	No subcontracting
Supplier origin	<code>prop_eu_bidders</code> , <code>prop_non_eu_bidders</code>	Domestic suppliers

Each variable is constructed as follows:

- **Supply type:** The variables `prop_services` and `prop_works` capture, respectively, the share of service and works contracts in total awarded tenders. The omitted category (*supplies*) includes contracts for the procurement of goods.
- **Award criterion:** `prop_meat` represents the share of contracts awarded on the basis of the *Most Economically Advantageous Tender* (MEAT) criterion, as opposed to the lowest-price rule.
- **Subcontracting:** `prop_subcontracted` measures the proportion of tenders explicitly allowing or involving subcontracting. This variable tests whether multi-tiered contracting arrangements are associated with fiscal expansion or efficiency gains.
- **Supplier origin:** `prop_eu_bidders` and `prop_non_eu_bidders` indicate the shares of contracts awarded to suppliers registered in other EU countries or outside the EU, respectively. The reference group includes contracts awarded to domestic suppliers, capturing the degree of internationalisation in procurement markets.

All variables are computed from the Global Public Procurement Dataset (GPPD) and directly reflect the distribution of awarded and winning bids. Unlike the main outsourcing indicator, these measures are not demeaned, as they represent meaningful proportions bounded between 0 and 1. Their inclusion in the extended specification allows testing whether compositional shifts in procurement activity, such as a greater reliance on services, quality-based criteria, or foreign suppliers, mediate the relationship between outsourcing and fiscal expenditure.

Demeaning of procurement composition controls To ensure consistency with the identification strategy developed in Chapter 4, all procurement composition variables are included in *demeaned* form in the econometric specifications. For each country i , the demeaned variable is constructed as:

$$prop_{x,it}^{dm} = prop_{x,it} - \overline{prop_{x,i}},$$

where $\overline{prop_{x,i}}$ denotes the country-specific time average of the corresponding procurement share.

This transformation removes all between-country differences in procurement structure, which typically reflect stable institutional, regulatory, or administrative features of national procurement systems rather than short-run policy fluctuations. Countries differ systematically in the extent to which they rely on services rather than goods, make use of MEAT criteria, contract out via subcontracting chains, or award tenders to foreign suppliers. Such permanent structural differences would otherwise enter the regression through the levels of the procurement composition variables and confound the interpretation of the estimated coefficients.

By demeaning each procurement composition variable, the model isolates *within-country temporal variation*—that is, deviations of procurement structure from a country’s own long-run average. Consequently, the estimated coefficients capture how *changes over time* in procurement composition (for example, an increase in the share of services, a temporary rise in MEAT awards, or a shift in the origin of suppliers) are associated with changes in public expenditure, net of all permanent cross-country differences.

This approach also ensures full coherence with the demeaned outsourcing measure X_{it} and the demeaned dependent variable $\ln spesa_{it}^{dm}$, aligning all key regressors with the within-country variation that underpins the two-way fixed-effects identification strategy.

3.2.3 Control Variables

A selected set of control variables is included in the empirical model to better isolate the association between outsourcing intensity and public spending. These variables account for macroeconomic fluctuations, demographic factors, fiscal stance, institutional quality, and political orientation. Except for bounded indices and dummy variables, all controls enter the specification in *logs of levels*. This approach (i) ensures consistency in scale with the dependent variable and (ii) allows the estimated coefficients to be interpreted as elasticities, thereby improving comparability across indicators.

Macroeconomic conditions: To capture cyclical variations in European economies, a *recession* dummy variable (Recession_{it}) is introduced. It takes the value one when country i experiences a year-on-year decline in real GDP in year t , and zero otherwise. This variable controls for cyclical contractions that typically trigger both automatic stabilisers and discretionary fiscal responses. Previous studies examining the determinants of public expenditure, such as Alonso et al. (2011) and Alonso et al. (2017), commonly include real GDP growth to capture business-cycle effects. Here, a binary recession indicator is preferred as a more robust and less endogenous proxy.

Labour market: $\ln(\text{Unemployment Rate}_{it})$ denotes the logarithm of the harmonised unemployment rate (as a percentage of the labour force). This variable captures variation in public expenditure arising from labour market dynamics. The unemployment rate is widely used in empirical research on fiscal dynamics and public spending as a control for cyclical labour market slack and welfare-state responsiveness.

Demographic structure: Population density, measured as the number of residents per square kilometre (in logarithm), captures both the intensity of demand for public services and the scale of fixed administrative costs. Although not included in the baseline specification for parsimony, the *dependency ratio*, measuring the share of dependents in the population, is frequently employed in related studies, as it is typically associated with greater welfare-state expenditure pressures (Alonso et al., 2017; Delgado-Téllez et al., 2022; Alonso et al., 2011).

Fiscal stance: $\ln(\text{Government Debt}_{it})$ represents general government gross debt as a percentage of GDP (source: Eurostat). This variable reflects the fiscal constraints faced by governments: higher debt levels generally impose consolidation pressures that may limit expenditure expansion. Including public debt as a control is standard in empirical analyses of fiscal behaviour, as it captures both the legacy of past fiscal policies and the intertemporal sustainability constraint of public finances. Moreover, controlling for debt helps to avoid omitted-variable bias arising from cross-country differences in fiscal space, as highlighted in comparative studies of European consolidation and expenditure adjustment.

Institutional quality: *Control of Corruption* (Worldwide Governance Indicators, World Bank) measures perceived corruption risk on a continuous scale ranging from -2.5 (weak control) to $+2.5$ (strong control). Procurement activities are among the most corruption-prone areas of public administration, and the GPPD dataset itself was originally designed to explore these dynamics (Fazekas and Tóth, 2021). Using a macro-level institutional indicator

such as WGI ensures consistency with cross-country analyses and mitigates measurement noise that would arise from micro-level corruption proxies.

Political orientation: *Government state–market orientation* (`gov_state_market`) is a continuous indicator derived from the *Manifesto Project Database*. It combines two policy dimensions—`planeco` (preference for economic planning) and `markeco` (support for market regulation)—aggregated across parties in the governing coalition using seat shares as weights. The resulting index ranges from negative (state-oriented, interventionist governments) to positive (market-oriented, liberal governments). This measure offers several advantages over the conventional left/right or party-family dummies frequently adopted in cross-country studies: (i) it captures the effective ideological balance of coalition governments, which dominate in the European political context; (ii) it varies continuously, allowing for more nuanced modelling of ideological gradients; and (iii) it directly measures the policy orientation relevant to marketisation and contracting-out decisions. This methodological refinement follows Abbott et al. (2025), who demonstrate that detailed, coalition-weighted indicators of government ideology explain contracting-out behaviour more effectively than binary political dummies. At the same time, the indicator’s reliance on party manifestos entails the limitation that it reflects stated preferences rather than realised policy actions, though this is mitigated by its annual updating and empirical validation in recent political-economy literature.

All control variables are constructed to be temporally coherent with the 2011–2021 GPPD window used for the key regressor. Dummies and indices (*recession*, *corruption control*, *state–market orientation*) enter in levels, while all other controls are log-transformed levels.

Table 3.4: Controls: definitions and transformations

Variable	Definition	Transformation
$\ln(\text{Unemployment Rate})$	Harmonised unemployment rate (Eurostat)	$\ln(\%)$
$\ln(\text{Population Density})$	Residents per km^2 (Eurostat)	$\ln(\text{level})$
$\ln(\text{Gov. Debt})$	General gov. gross debt / GDP (Eurostat)	$\ln(\%)$
<i>Control of Corruption</i>	WGI index (−2.5 to +2.5)	raw index
<i>Gov. state–market</i>	Seat-weighted cabinet ideology (Manifesto)	raw index
<i>Recession</i>	1 if real GDP growth < 0 (Eurostat), 0 otherwise	dummy (0/1)

Comprehensive metadata describing all variables used in the empirical analysis, together with their sources, units of measurement, and transformation procedures, are presented in Appendix B. This supplementary material ensures full reproducibility and facilitates cross-country comparison of indicators.

3.3 Descriptive Statistics

This section documents the main empirical patterns of the dataset used in the analysis. We begin by presenting summary statistics for the key variables and by decomposing their variance into between-country and within-country components, clarifying the structure of the panel. We then examine temporal dynamics over 2011–2021 and describe cross-country heterogeneity in expenditure and outsourcing intensity. These descriptive patterns provide the empirical background for the analysis developed in the following sections.

3.3.1 Summary Statistics

The descriptive statistics for all variables included in the country–year panel (231 observations) are reported in Table 3.5. Both the dependent variable, demeaned log real primary expenditure ($\ln(\text{primary exp.})_{it}^{dm}$), and the demeaned outsourcing indicator (X_{it}) display means that are, by construction, approximately zero (-0.00006 and -3.07×10^{-17} respectively). Their dispersion reflects only within–country temporal variation, with standard deviations equal to 0.117 for $\ln(\text{primary exp.})_{it}^{dm}$ and 1.19 for X_{it} .

The raw log mean contract value ($\ln(\text{valmean})$) exhibits substantial heterogeneity across member states, ranging from 9.07 to 18.2 with a mean of 14.5 and a standard deviation of 1.70. The macroeconomic controls display variability consistent with known structural differences within the EU: population density varies between 2.89 and 7.39 in logs (sd 0.877), while government debt ranges from 2.14 to 5.34 (sd 0.614). Institutional indicators show similarly wide cross-country dispersion: *control of corruption* spans from -0.381 to 2.40 , and the cabinet *state–market orientation* index ranges from -8.62 to 12.5 . Finally, the recession dummy equals one in roughly 17% of the country–year observations, indicating the presence of negative-growth episodes during the 2011–2021 period.

Table 3.5: Summary statistics of main variables (country–year panel, demeaned logs)

Variable	Mean	Median	SD	Min	Max
$\tilde{\ln}(\text{Primary Expenditure})$	0.000	−0.019	0.117	−0.250	0.329
$\ln(\text{Mean Contract Value})$	14.52	14.70	1.70	9.07	18.20
$\tilde{\ln}(\text{Outsourcing Intensity})$	0.000	0.094	1.19	−4.47	4.03
$\ln(\text{Unemployment Rate})$	1.99	1.96	0.46	0.693	3.28
$\ln(\text{Population Density})$	4.67	4.69	0.877	2.89	7.39
$\ln(\text{Government Debt/GDP})$	4.06	4.10	0.614	2.14	5.34
Control of Corruption (index)	0.947	0.742	0.801	−0.381	2.40
State–Market Orientation (index)	−1.16	−0.487	3.31	−8.62	12.50
Recession (dummy)	0.169	0.000	0.375	0	1

Notes: Statistics computed over 231 country–year observations (27 EU member states, 2011–2021). Italy 2016 is excluded due to anomalous contract-value reporting. Tildes denote variables demeaned within countries over the full sample period.

3.3.2 Variance Decomposition

Table 3.6 reports a variance decomposition for the main variables of interest in the 2011–2021 balanced panel.

The dependent variable used in the empirical section is the demeaned log level of real primary expenditure,

$$\ln(\text{expenditure}_{it})^{dm} = \ln(\text{expenditure}_{it}) - \overline{\ln(\text{expenditure}_i)},$$

which removes all time-invariant cross-country heterogeneity. As expected, its variance is entirely driven by within-country fluctuations, with a between-country component equal to zero. The same holds by construction for the demeaned outsourcing indicator,

$$X_{it} = \ln(\text{mean contract value}_{it}) - \overline{\ln(\text{mean contract value}_i)}.$$

The macroeconomic controls exhibit markedly different patterns. Variables such as population density, government debt, and control of corruption are overwhelmingly cross-sectional, displaying between-country shares above 96–99%. Conversely, the recession dummy is almost entirely driven by within-country variation (94%), reflecting its cyclical nature. Political orientation and unemployment show a more balanced structure, with 32–35% of their variance attributable to within-country changes.

Overall, these results indicate that the panel is characterised by strong structural heterogeneity across EU member states, while only a subset of covariates contains economically

meaningful variation over time. This informs the identification strategy adopted in the subsequent sections.

Table 3.6: Variance decomposition of key variables (2011–2021)

Variable	Between share	Within share	Total variance
Demeaned log primary expenditure ($\ln \exp_{it}^{dm}$)	0.00	1.00	0.0155
Demeaned outsourcing indicator (X_{it})	0.00	1.00	1.5300
Recession dummy	0.059	0.941	0.160
Government state–market orientation	0.652	0.348	11.200
Log unemployment rate	0.682	0.318	0.220
Log government debt	0.965	0.035	0.389
Control of corruption	0.979	0.021	0.629
Log population density	0.999	0.001	0.878

3.3.3 Temporal Patterns (2011–2021)

This section documents the evolution of real primary expenditure and outsourcing intensity across EU–27 member states between 2011 and 2021. Both series are demeaned at the country level, such that values represent deviations from a country’s long-term average. Positive values therefore indicate above-trend expenditure (or outsourcing), while negative values reflect below-trend levels.

Figure 3.1 shows that, on average, real primary expenditure remains persistently below its country-specific trend in the aftermath of the European sovereign debt crisis. The series stabilises around zero between 2016 and 2018, consistent with the gradual relaxation of fiscal consolidation rules, and turns positive in 2020–2021. The upward deviation in these final years aligns with the COVID-19 shock, during which fiscal policy became strongly expansionary and public expenditure increased across member states.

The demeaned outsourcing indicator, plotted in Figure 3.2, displays a distinct pattern. Outsourcing remained below its long-run country average until 2016 but shifted to systematically positive deviations from 2017 onward. This change indicates a sustained increase in the relative importance of outsourced procurement contracts in the post-crisis period. The pronounced rise in 2020 is consistent with the reallocation of public expenditure during the pandemic.

Overall, the two temporal profiles highlight that (i) fiscal consolidation after 2008 suppressed real expenditure across member states, (ii) outsourcing began increasing earlier than public expenditure itself, and (iii) both indicators rise markedly during the pandemic years.

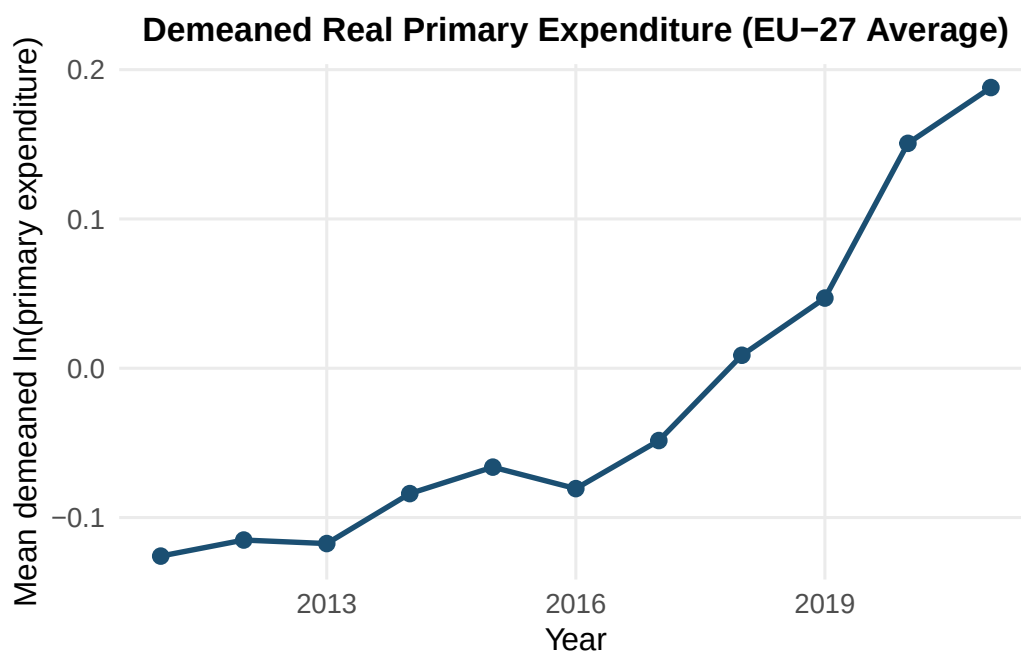


Figure 3.1: Demeaned real primary expenditure, EU-27 average (2011–2021).

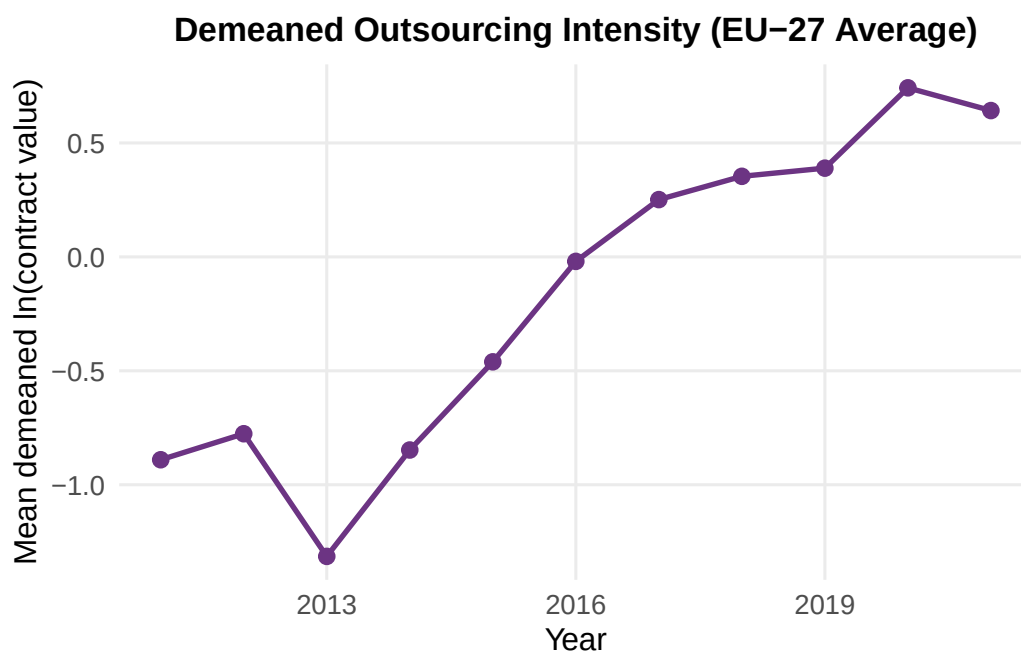


Figure 3.2: Demeaned outsourcing intensity, EU-27 average (2011–2021).

These stylised facts motivate the empirical analysis that follows, which tests whether deviations in outsourcing intensity are systematically associated with deviations in real primary expenditure.

3.3.4 Cross-Country Heterogeneity

Table 3.7 summarises the within-country dispersion of the demeaned variables over the 2011–2021 period. For each Member State, the reported standard deviations capture the amplitude of year-to-year fluctuations in real primary expenditure and outsourcing intensity around national long-run averages, thereby isolating the temporal volatility of fiscal and procurement dynamics net of structural level differences.

The data reveal a relatively contained degree of volatility in demeaned primary expenditure across most countries, with standard deviations typically below 0.10 in several Northern and Western European cases (e.g. DK 0.046, FI 0.047, FR 0.050). In contrast, a subset of Eastern and Southern Member States exhibits more pronounced fluctuations (e.g. RO 0.198, LT 0.188, MT 0.207), suggesting more variable spending trajectories over the sample period.

Outsourcing intensity displays substantially larger and more heterogeneous volatility. While some countries show limited dispersion (e.g. MT 0.389, CY 0.421, PL 0.425), others experience markedly higher fluctuations (e.g. NL 2.15, LV 2.24, AT 1.87). Notably, the cross-country pattern does not align closely with expenditure volatility: countries with relatively stable expenditure profiles may nonetheless exhibit highly volatile outsourcing dynamics (e.g. NL, AT), indicating that the procurement cycle evolves with greater amplitude than aggregate spending. This heterogeneity supports the relevance of modelling within-country temporal variation and underscores the need for econometric specifications capable of accommodating differentiated national trajectories.

Table 3.7: Within-country volatility of demeaned variables (2011–2021)

Country	SD(ln expenditure ^{dm})	SD(Outsourcing X)
LV	0.150	2.240
SK	0.116	0.748
SI	–	–
RO	0.198	1.430
PL	0.146	0.425
LT	0.188	1.540
EE	0.165	1.390
BG	0.150	1.350
HR	0.105	0.464
CZ	0.141	1.350
HU	0.136	1.270
SE	0.052	1.270
IE	0.143	1.130
FI	0.047	1.120
DK	0.046	0.878
ES	0.097	1.470
PT	0.066	0.797
EL	0.079	1.090
IT	0.075	0.631
CY	0.158	0.421
MT	0.207	0.389
FR	0.050	1.040
LU	0.124	1.150
AT	0.096	1.870
DE	0.112	0.657
BE	0.075	1.220
NL	0.080	2.150

Notes: Volatility is measured as the standard deviation of each demeaned series over 2011–2021. Demeaning is performed at the country level to remove structural cross-sectional differences. “–” denotes insufficient time variation due to missing observations.

3.4 Preliminary Insights

This section provides a descriptive assessment of the association between demeaned outsourcing intensity and demeaned real primary expenditure. Since both variables are demeaned at the country level, the correlations reported below capture purely within-country co-movement over time, net of structural cross-country differences.

Table 3.8 documents the within-country correlation between the two variables for each Member State. The results reveal a clear pattern of heterogeneity. Several Eastern European countries display strong positive co-movement (e.g. HU 0.855, EE 0.820, RO 0.770, LT 0.715), indicating that years of above-trend outsourcing tend to coincide with above-trend expenditure. A number of Western and Northern European countries also exhibit positive, though more moderate, correlations (e.g. FR 0.669, AT 0.626, IE 0.571). Other Member States show weaker associations (e.g. DK, FI, DE) or correlations close to zero (ES). A small subset of countries presents negative co-movement (EL, HR, SK), suggesting that outsourcing and expenditure sometimes evolve in opposite directions, potentially reflecting the effects of fiscal consolidation episodes.

To complement these findings, Table 3.8 reports the cross-sectional correlation between the two demeaned variables for each year. These correlations are generally small and unstable over time, rarely exceeding 0.15 except in 2011 and during the pandemic years 2020–2021. This indicates that the association between outsourcing and expenditure is not driven by cross-country differences in average levels, but rather by temporal dynamics within countries.

Taken together, these preliminary insights show that the relationship between outsourcing and primary expenditure is heterogeneous across Member States and primarily operates along the time dimension. This underscores the relevance of modelling within-country variation and motivates the panel regression framework introduced in the next section.

Table 3.8: Correlation between demeaned outsourcing intensity and demeaned primary expenditure

Panel A: Within-country correlation

Country	Corr	N
HU	0.855	11
EE	0.820	9
RO	0.770	11
LT	0.715	10
PL	0.672	8
FR	0.669	9
LV	0.648	8
CY	0.630	7
AT	0.626	11
IT	0.572	9
IE	0.571	9
BG	0.525	9
BE	0.518	8
CZ	0.433	10
MT	0.422	6
SE	0.406	8
NL	0.395	9
DK	0.251	11
LU	0.244	8
PT	0.197	8
FI	0.196	8
DE	0.184	11
ES	0.076	9
EL	-0.385	8
HR	-0.515	6
SK	-0.645	9
SI	—	1

Panel B: Within-year (cross-sectional) correlation

Year	Corr	N
2011	0.514	5
2012	0.139	9
2013	0.076	16
2014	-0.049	21
2015	0.064	24
2016	0.080	26
2017	0.153	26
2018	0.121	26
2019	0.167	26
2020	0.345	26
2021	0.309	26

Notes: Correlations computed using demeaned variables. Panel A reports within-country temporal correlations; Panel B reports cross-sectional correlations by year. “—” indicates insufficient variation.

Chapter 4

Methodology

This chapter develops the econometric framework used to analyse the relationship between outsourcing intensity and real primary expenditure. The aim is to formalise the empirical model, clarify its identifying assumptions, and discuss the specification choices required by the structure of the data.

The chapter proceeds as follows: Section 4.1 introduces the empirical model and defines the variables entering the specification; Section 4.2 motivates the econometric approach by examining key features of the panel and presenting formal model selection tests; Section 4.3 assesses whether a dynamic structure is required and reports diagnostic checks on persistence; Section 4.4 concludes by outlining the implications for the identification strategy adopted in the empirical analysis.

4.1 Econometric Specification

The empirical analysis examines whether changes in outsourcing intensity are systematically associated with changes in real primary expenditure within EU member states over time. To focus on within-country dynamics, the dependent variable and the main explanatory variable are expressed in natural logarithms and demeaned at the country level. All other continuous controls enter in logs but are not demeaned, while institutional indices and the recession dummy enter in levels.

4.1.1 Baseline Model

Let $\tilde{y}_{it}$ denote demeaned log real primary expenditure for country i in year t , and let X_{it} denote demeaned log outsourcing intensity. The baseline empirical model is:

$$\tilde{y}_{it} = \beta X_{it} + \gamma' Z_{it} + \alpha_i + \lambda_t + \varepsilon_{it}, \quad (4.1)$$

where α_i are country fixed effects capturing time-invariant heterogeneity, and λ_t are year fixed effects absorbing common shocks to all member states. The coefficient β measures the association between deviations in outsourcing and deviations in primary expenditure relative to each country's long-run average. The vector Z_{it} includes macroeconomic, demographic and institutional controls: log unemployment, log population density, log government debt, the recession indicator, the control-of-corruption index, and the state-market orientation of the cabinet.

Since $\tilde{y}_{it}$ and X_{it} are demeaned by construction, identification in equation (4.1) relies entirely on within-country temporal variation. This specification corresponds to the standard additive two-way fixed-effects estimator.

4.1.2 Error Structure and Inference

Diagnostic tests reported in Appendix C indicate the presence of heteroskedasticity, serial correlation, and moderate cross-sectional dependence in the residuals. To obtain consistent inference under these forms of misspecification, the baseline estimates are reported with heteroskedasticity- and autocorrelation-robust standard errors clustered at the country level. This correction is standard in macro-panel settings where N is moderate and T is relatively small.

4.1.3 Interpretation of Coefficients

Given the log-linear structure of the model, β has the interpretation of a within-country elasticity: it captures the percentage deviation in real primary expenditure associated with a one percent deviation in outsourcing intensity around each country's historical average. The coefficients γ on the control variables measure conditional elasticities with respect to macroeconomic and institutional covariates, while the fixed effects ensure that identification is not affected by structural cross-country differences or year-specific shocks.

Overall, the specification in equation (4.1) provides a transparent and internally consistent framework for analysing the co-movement between outsourcing intensity and real primary expenditure over time.

4.2 Justification for the Empirical Strategy

4.2.1 Evidence from the Structure of the Data

The empirical properties of the dataset inform the choice of a two-way fixed-effects specification. As documented in the variance decomposition reported in Section 3.3.2, both the demeaned dependent variable, log real primary expenditure, and the demeaned outsourcing indicator exhibit a between-country variance component equal to zero. This follows directly from the transformation applied to both series, which removes all time-invariant cross-sectional differences. Consequently, the identifying variation is exclusively within-country and arises from temporal deviations around each country’s mean.

The control variables display heterogeneous patterns of variation. Population density, government debt, and control of corruption have between-country variance shares exceeding 96–99%, indicating that these indicators primarily reflect structural cross-sectional differences. Other variables, such as unemployment and the government’s state–market orientation, present a more balanced distribution of between and within variation. The recession indicator is almost entirely driven by within-country changes, consistent with its interpretation as a cyclical variable.

These characteristics have direct implications for the empirical design. Since the dependent variable and the main regressor contain only within-country variation, the estimator must rely on temporal changes to achieve identification. At the same time, the presence of substantial and persistent cross-country heterogeneity across several covariates motivates the inclusion of country-specific intercepts. Second, the sample period includes well-known macroeconomic shocks, most notably the sovereign debt crisis (2011–2014) and the COVID-19 pandemic (2020–2021), that affect all EU member states simultaneously. Allowing for a full set of year-specific intercepts therefore accommodates common shocks that would otherwise be confounded with the regressors of interest.

Overall, the variance decomposition indicates that the panel combines high cross-sectional heterogeneity with limited but economically relevant within-country temporal variation. Under these conditions, a two-way fixed-effects specification is consistent with the statistical properties of the data and with the transformations applied in the empirical analysis.

4.2.2 Formal Specification Tests

The adequacy of the two-way fixed-effects estimator is evaluated through a set of standard specification tests. These tests determine whether country and time intercepts are statistically required and whether the assumptions associated with alternative estimators, such

as pooled OLS or random effects, are consistent with the empirical properties of the data. Detailed results are reported in Appendix C.

Fixed effects versus pooled OLS. F-tests for the joint significance of the country and time intercepts reject the null of homogeneous intercepts across both dimensions. The country fixed-effects test ($F(26, 197) < 0.001$) indicates that unobserved heterogeneity across member states is systematic and cannot be captured by a pooled model. The time fixed-effects test ($F(10, 213) < 0.001$) shows that year-specific shocks are statistically relevant. These results rule out the pooled OLS specification and justify the use of fixed effects along both the cross-sectional and temporal dimensions.

Fixed effects versus random effects. The Hausman specification test evaluates whether the random-effects estimator is consistent under the null hypothesis that the regressors are uncorrelated with the unit-specific unobservables. The test rejects the null ($p < 0.001$), implying that the random-effects assumption is not supported by the data. In this setting, random effects would therefore be inconsistent, and the fixed-effects estimator remains the appropriate choice.

Variance component in the random-effects model. The Breusch–Pagan Lagrange Multiplier test assesses whether the variance of the unit-specific random effect is statistically different from zero. The test does not reject the null, consistent with an estimated variance component close to zero. While this result indicates that the random-effects model does not provide a statistically distinct alternative to pooled OLS, it does not imply that random effects are preferable; the Hausman test confirms that their identifying assumptions are not compatible with the data.

Taken together, the specification tests provide consistent evidence in favour of the two-way fixed-effects estimator. Both cross-sectional and temporal heterogeneity are empirically relevant, and the assumptions required for a random-effects specification are not supported. The fixed-effects framework adopted in the empirical analysis is therefore aligned with the statistical properties of the panel.

4.2.3 Methodological Positioning within the Existing Literature

The empirical literature on outsourcing and public expenditure uses different panel methods, reflecting substantial differences in research questions, data structures and temporal coverage. This section situates the econometric strategy adopted in this dissertation within

this methodological landscape, highlighting points of convergence and divergence relative to existing approaches.

A first distinction concerns the nature of the research questions addressed in prior work. Cross-country studies such as Alonso et al. (2011) and Alonso et al. (2017) investigate whether long-term reforms in public management (outsourcing, decentralisation, privatisation) affect the size and composition of the public sector over horizons of 20–30 years. Their focus on long-run fiscal dynamics motivates the use of specifications capable of distinguishing short- and long-run relationships, including panel DOLS and ARDL error-correction models. These techniques are appropriate in the presence of persistent series, potential non-stationarity and slow-moving institutional variables, but they require long time spans and meaningful low-frequency variation.

The present study pursues a different empirical objective. Rather than estimating long-run fiscal adjustment paths, it examines whether deviations in outsourcing intensity are associated with short- to medium-run deviations in real primary expenditure within each country over the period 2011–2021. Identification therefore relies on relatively high-frequency within-country movements, which are informative even in short panels. The econometric structure must consequently prioritise flexibility in the cross-sectional dimension and robustness to heterogeneous country-specific trends, rather than modelling long-run equilibria.

These differences in research focus align with important differences in data structure. The panels used in Alonso et al. (2011) and Alonso et al. (2017) cover European and OECD countries over roughly 20–30 years, allowing the exploitation of long-term dynamics and gradual institutional changes. By contrast, the dataset used in this dissertation spans a shorter horizon (2011–2021) and relies on demeaned logarithmic transformations for both the dependent variable and the main regressor. As a result, the identifying variation is entirely temporal and specifically centred on deviations from country-specific means. This construction removes all cross-sectional level differences, leaving a structure that naturally aligns with a static two-way fixed-effects estimator.

A second point of comparison concerns the level of analysis. Regional studies, such as Delgado-Téllez et al. (2022) for the Spanish regions, work with panels in which the time dimension is moderate (around sixteen annual observations) and key variables such as spending and outsourcing display substantial persistence. In this context, the authors adopt an Anderson–Hsiao panel IV estimator, instrumenting lagged spending and outsourcing with their second lags. This approach is suitable for panels with limited time depth, in which more demanding GMM estimators would suffer from instrument proliferation and weak identification. Micro-level analyses such as Barkley (2021) use project-level data to study cost and efficiency implications of outsourcing, typically estimating models with agency or project

fixed effects and rich sets of contract characteristics. These designs differ from the macro-panel setting considered here, both in terms of the outcome of interest and in the structure of the available variation.

Across these strands of the literature, a unifying feature is the systematic use of fixed effects to control for unobserved heterogeneity, whether at the country, regional or agency level. In this sense, the two-way fixed-effects approach adopted in this dissertation is consistent with the identification logic underpinning both macro-level and micro-level work: the key variation is within-unit, not cross-sectional.

Taken together, the comparison with existing empirical studies indicates that the two-way fixed-effects specification used in this dissertation is consistent with prevailing methodological practice, while appropriately tailored to the specific structure of the data. The limited time dimension and the demeaned design of the key variables favour a static TWFE estimator over dynamic or cointegration-based methods, which are better suited to longer time series or granular administrative datasets. Given the moderate persistence of demeaned primary expenditure documented in Appendix C, it is natural to ask whether a dynamic extension of the baseline specification is nonetheless warranted; the next section addresses this issue.

4.3 Investigating Dynamic Specification

A further methodological consideration concerns the potential role of dynamics in the relationship between outsourcing intensity and primary expenditure. In empirical settings where the outcome variable adjusts gradually over time, omitting a lagged dependent variable may lead to misspecification or to an incomplete representation of short-run adjustment mechanisms. This section examines whether such a dynamic extension is empirically warranted in the present context and whether it can be estimated with sufficient reliability given the structure of the data.

Temporal dependence in demeaned primary expenditure. Appendix C reports country-specific lag-1 autocorrelation coefficients for the demeaned dependent variable. Although demeaning removes all time-invariant heterogeneity across countries, the resulting series still display non-trivial temporal dependence, with several coefficients above 0.6. This pattern indicates that annual changes in primary expenditure are not purely idiosyncratic and motivates an exploratory assessment of whether a dynamic panel specification could capture additional adjustment behaviour.

Dynamic fixed-effects estimation. To evaluate this possibility, the baseline two-way fixed-effects model was augmented with a lagged dependent variable. Across a sequence of

nested specifications, the autoregressive coefficient remains large and precisely estimated, consistent with the autocorrelation evidence.

Feasibility of dynamic panel estimators. While the dynamic FE estimates offer useful descriptive insight, their interpretation is limited by the well-known Nickell bias, which affects fixed-effects estimators whenever the time dimension is short. After demeaning, the effective time span of the panel is approximately seven to nine observations per country, a range in which the bias in $\hat{\rho}$ is non-negligible and cannot be corrected through large- T approximations.

To examine whether this concern also reflects endogeneity of the lagged dependent variable, an Anderson–Hsiao estimator was implemented using $\tilde{y}_{i,t-2}$ as an instrument. In short panels, this is effectively the only feasible IV strategy: more general difference or system GMM estimators would require a larger time dimension to prevent instrument proliferation and weak-instrument pathologies. In this application, the IV estimate of the autoregressive coefficient is very close to the fixed-effects estimate, and a Hausman-type comparison does not reject their equality (Appendix C). This suggests that, given the available sample, the potential endogeneity of the lagged dependent variable is not empirically detectable.

Implications for model choice. Taken together, the empirical evidence points in a clear and internally consistent direction. The diagnostics indicate that real primary expenditure displays a degree of persistence that is fully compatible with a dynamic specification, and the exploratory estimations confirm that the autoregressive component is empirically relevant. The Anderson–Hsiao instrumentation further suggests that the inclusion of a lagged dependent variable is econometrically feasible: the IV and fixed-effects estimates of the autoregressive term are very similar, and no detectable endogeneity of the lagged dependent variable emerges in this sample. In this sense, a dynamic panel model constitutes a conceptually valid extension of the baseline framework.

At the same time, the structure of the dataset places substantive constraints on the reliable estimation of such a model. The short time dimension, combined with the use of demeaned transformations, limits the amount of within-country temporal variation available to identify dynamic effects. In this setting, conventional estimators for dynamic panels are susceptible to small- T distortions, and the Anderson–Hsiao procedure, while technically appropriate, does not provide a sufficiently robust basis for drawing substantive conclusions on dynamic adjustment mechanisms.

For these reasons, and in light of the preliminary and exploratory nature of this study, the empirical analysis proceeds with the static two-way fixed-effects specification introduced in Section 4.1. A more comprehensive investigation of dynamic behaviour—potentially using

longer time series or more granular administrative data—remains a promising direction for future research and a natural complement to the static framework adopted here.

4.4 Summary

This chapter has established the econometric foundations of the empirical analysis by deriving the specification consistent with the structure of the data and the identification requirements of the research question. The demeaned construction of the key variables confines identification to within-country variation, and the variance decomposition confirms that this is the only dimension of the data that carries economically meaningful information for the relationship of interest. Formal specification tests further demonstrate that unobserved heterogeneity across countries and time is empirically relevant, thereby validating the two-way fixed-effects framework as the appropriate baseline model.

The methodological discussion is anchored in the broader empirical literature, but also highlights the distinctive features of the present dataset—namely the short time horizon and the reliance on within-unit transformations—which differentiate this study from long-run analyses of institutional reforms. Within this context, the static two-way fixed-effects estimator emerges as the specification that most efficiently exploits the available variation while maintaining robustness to cross-sectional and temporal confounders.

The chapter has also assessed the scope for dynamic modelling. Persistence diagnostics and exploratory estimations indicate that adjustment dynamics are plausible from an economic standpoint; however, the limited time dimension of the panel and the implications of small-T bias restrict the feasibility of reliably estimating a dynamic specification. The exploratory analysis thus clarifies that dynamic behaviour may be relevant in principle, but cannot be rigorously identified within the constraints of the dataset. Consequently, and in line with the preliminary and exploratory nature of this study, the empirical analysis proceeds with the static two-way fixed-effects model, while recognising that dynamic extensions constitute a promising direction for future research.

Chapter 5

Results and Discussion

5.1 Introduction

This chapter presents the empirical results of the analysis and addresses the research question formulated in Chapter 2, which concerns whether short-run changes in outsourcing are associated with corresponding movements in real primary public expenditure across EU Member States.

The empirical strategy, outlined in Chapter 4, employs a two-way fixed-effects framework estimated on a country–year panel from 2011 to 2021. The specification exploits within-country variation over time while absorbing time-invariant cross-country heterogeneity and common shocks, and includes a set of macro-fiscal and institutional controls.

The remainder of the chapter mirrors the structure of the analysis. The section on *Empirical Results* reports the baseline estimates, incorporates procurement composition variables to assess robustness, and examines the decomposition of public expenditure across economic and functional categories. The *Discussion* section interprets the findings in light of the theoretical mechanisms and empirical literature and reviews the main limitations related to data structure, identification, endogeneity, and stationarity. A brief summary concludes the chapter.

5.2 Empirical results

5.2.1 Baseline TWFE Estimates

Table 5.1 reports the baseline two-way fixed-effects estimates of the relationship between outsourcing intensity and real primary public expenditure. The coefficient on the demeaned logarithm of outsourcing intensity is positive and statistically significant at the 5% level ($\hat{\beta} = 0.0073$). Because both the dependent variable and the variable of interest are expressed in demeaned logarithms, the coefficient represents a *within-country elasticity*: a temporary 1% increase in outsourcing intensity relative to a country’s long-run mean is associated with an increase of approximately 0.007% in real primary expenditure relative to its own country-specific mean. Interpreting the magnitude of the elasticity requires recognising that the specification captures only short-run deviations around smooth and persistent national expenditure paths; as documented in Chapter 4, real primary expenditure exhibits substantial autoregressive behaviour, so immediate responses to changes in outsourcing are necessarily limited in size.

Identification derives exclusively from within-country variation over time, as country and year fixed effects absorb all time-invariant heterogeneity across Member States and any shocks common to all countries. The resulting estimate should therefore be interpreted as a contemporaneous reduced-form elasticity rather than as evidence of structural long-run effects. Standard errors are clustered at the country level to allow for arbitrary heteroskedasticity and serial correlation within countries, an essential correction in short macroeconomic panels.

Among the macro-fiscal controls, only the unemployment rate is statistically significant. Its negative coefficient is consistent with counter-cyclical fiscal adjustments, whereby periods of labour-market slack tend to coincide with temporary reductions in primary expenditure. The remaining controls—population density, government debt, corruption control, and state-market governance—display no detectable within-country effects once fixed effects are included. This outcome is expected for covariates that evolve slowly or with limited short-run variability: after absorbing structural differences through fixed effects, little residual variation remains available for identification.

The goodness-of-fit measures reflect these properties. The overall R^2 (0.832) is high because country fixed effects capture most of the cross-sectional variation in expenditure levels. By contrast, the within R^2 (0.023) indicates that only a modest share of short-run deviations from country-specific expenditure trends is explained by outsourcing intensity and the included covariates, which is typical of highly aggregated and persistent fiscal variables. The adjusted R^2 (0.827) further confirms that most explanatory power arises from the fixed

effects rather than from contemporaneous regressors.

Taken together, the baseline estimates reveal a statistically detectable but quantitatively limited short-run elasticity between outsourcing intensity and real primary public expenditure. While the association is positive and statistically significant, its magnitude is modest, which is consistent with the high persistence of the dependent variable and the reduced-form nature of the specification. Section 5.2.2 evaluates whether this relationship is sensitive to the inclusion of procurement composition controls.

Table 5.1: Baseline TWFE estimates: outsourcing and real primary expenditure

Variable	Estimate	Std. Error	p-value
Outsourcing intensity (X)	0.00733**	0.00314	0.0207
Unemployment rate (ln)	-0.1109**	0.04368	0.0120
Population density (ln)	0.2704	0.48068	0.5744
Government debt (ln)	0.07763	0.05375	0.1503
Corruption control	0.03099	0.04953	0.5323
State-market governance	0.00325	0.00352	0.3575
Recession indicator	0.03911	0.02475	0.1157
<i>Observations</i>		231	
<i>Countries</i>		27	
R^2 (overall)		0.8323	
R^2 (adjusted)		0.8271	
R^2 (within)		0.0225	

Notes: Dependent variable: demeaned log real primary expenditure. Two-way fixed-effects estimator with country and year fixed effects. Standard errors clustered at the country level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2.2 Robustness: Procurement Controls

Table 5.2 extends the baseline specification by including demeaned indicators of procurement composition. These variables capture within-country reallocations across supply types, award criteria, subcontracting arrangements, and supplier origin. The coefficient on outsourcing intensity remains positive and statistically significant ($\hat{\beta} = 0.0065$), with a magnitude very close to the baseline estimate, indicating that the elasticity is not driven by short-run changes in procurement mix.

Among procurement composition variables, two patterns emerge. First, years in which a greater share of contracts involve subcontracting—relative to contracts without subcontracting—are associated with higher primary expenditure. Second, reallocations in supplier origin display a systematic association: a higher share of contracts awarded to non-EU suppliers—relative to domestic firms—is correlated with higher primary expenditure, while the analogous share of EU suppliers shows a weaker pattern of the same sign.

The negative coefficient on service contracts—defined relative to goods—is marginally significant and merits clarification. In the GPPD data, service contracts are on average considerably smaller in value than goods contracts, which implies that an increase in the share of services tends to shift procurement activity toward lower-value procedures. This provides a descriptive rationale for the negative sign but does not identify the underlying mechanism. The reduced-form evidence cannot distinguish whether the coefficient reflects differences in contract size, in the composition of procured services, or in sector-specific patterns within the broad service category. A more granular classification of services would be necessary to disentangle these factors.

The remaining procurement controls do not display measurable within-country effects. Reallocations from goods to works contracts and changes in the incidence of MEAT awards show no significant correlations with expenditure once fixed effects are absorbed.

Including procurement composition indicators produces only a modest increase in model fit: the overall R^2 rises from 0.832 to 0.840, and the within R^2 from 0.023 to 0.031. This indicates that procurement mix variables account for only a small share of the residual within-country variation in demeaned primary expenditure. Most explanatory power continues to be provided by the fixed effects, which absorb structural cross-country differences and common time shocks. Accordingly, the inclusion of procurement composition does not materially alter the estimated elasticity between outsourcing intensity and primary expenditure.

Table 5.2: Two-way fixed-effects estimates with procurement composition controls

Variable	Estimate	Std. Error
X (outsourcing intensity)	0.00646 **	0.00315
prop_services _{dm}	-0.03276 *	0.01953
prop_works _{dm}	-0.00319	0.02896
prop_meat _{dm}	-0.01475	0.01400
prop_subcontracted _{dm}	0.05376 **	0.02583
prop_eu_bidders _{dm}	0.04858 *	0.02869
prop_non_eu_bidders _{dm}	0.07682 **	0.03061
ln(unemployment rate)	-0.10126 **	0.04619
ln(population density)	0.37997	0.46966
ln(government debt)	0.05835	0.06056
corruption control	0.03189	0.04727
state-market governance	0.00353	0.00363
recession	0.04932 **	0.02492
Observations	231	
Countries	27	
R² overall	0.840	
R² within	0.031	
Adjusted R² overall	0.831	

Notes: Two-way fixed-effects estimates with country and year fixed effects. All procurement composition variables are demeaned at the country level. Standard errors clustered at the country level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2.3 Decomposition of Public Expenditure

This subsection examines whether the elasticity identified in the baseline model is reflected in specific components of public expenditure. Two complementary decompositions are considered. The first follows the ESA 2010 economic classification, focusing on intermediate consumption (P.2), social transfers in kind via market producers (D.1), and gross fixed capital formation (P.5). The second employs the functional COFOG framework, assessing whether outsourcing intensity co-moves with major spending functions such as general public services (TF6_1), public order and safety (TF6_3), economic affairs (TF6_4), health (TF6_7), education (TF6_9), and social protection (TF6_10).

Tables 5.3 and 5.4 report the estimated coefficients. Across all specifications, the coefficient on outsourcing intensity remains positive but is never statistically significant, and the associated standard errors are markedly larger than in the aggregate model. This pattern reflects the very limited within-country volatility of disaggregated expenditure categories: once demeaned at the country level, these series display little short-run variation, and the two-way fixed-effects transformation absorbs most of their remaining temporal structure. As a consequence, the model lacks sufficient within-country signal to detect systematic short-run associations between outsourcing and specific expenditure components.

The absence of statistically detectable effects is consistent with the persistence and institutional rigidity of the decomposed aggregates. Capital expenditure, social transfers in kind, and sectoral COFOG functions evolve slowly from year to year and respond to policy and macroeconomic conditions only with considerable inertia. In this setting, short-run fluctuations in outsourcing intensity are unlikely to coincide with equally rapid adjustments in any single category. Importantly, the coefficient on outsourcing intensity retains a positive sign in all models, although the confidence intervals are wide and do not allow any sector-specific interpretation.

Finally, it is important to emphasise the exploratory nature of these decompositions. Their purpose is not to identify precise channels but to document whether the aggregate elasticity masks heterogeneous sectoral responses. The evidence does not suggest such heterogeneity: the baseline association does not appear to operate through identifiable movements in individual components of public expenditure, which is consistent with the limited temporal variation and structural persistence of these series. It is nevertheless plausible that outsourcing affects narrower sub-components of these aggregates, such as specific goods and services items within COFOG functions, that cannot be detected using the available functional and economic classifications.

Table 5.3: Economic decomposition: intermediate consumption (P.2), social transfers in kind (D.1), and capital formation (P.5)

	P.2	D.1	P.5
Outsourcing (X)	0.0169 (0.0969)	0.0159 (0.0960)	0.0277 (0.1009)
ln unemployment rate	0.3668 (0.8348)	0.3370 (0.8448)	0.3037 (0.8415)
ln population density	7.7681 (10.4298)	6.5578 (10.3084)	8.4445 (10.7639)
ln government debt	0.0802 (1.3566)	0.2207 (1.3430)	-0.0506 (1.4283)
corruption control	-0.0855 (1.1184)	0.0854 (1.1497)	-0.1822 (1.1042)
state-market governance	-0.0705 (0.0372)	-0.0692 (0.0366)	-0.0681 (0.0368)
recession	-0.7025 (0.5427)	-0.6663 (0.5329)	-0.7238 (0.5461)
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: Standard errors clustered at the country level in parentheses. $^*p < 0.10$.

Table 5.4: Functional decomposition (COFOG, Eurostat TF6_x): general public services (TF6_1), public order and safety (TF6_3), economic affairs (TF6_4), health (TF6_7), education (TF6_9), and social protection (TF6_10)

	TF6_1	TF6_3	TF6_4	TF6_7	TF6_9	TF6_10
Outsourcing (X)	0.0186 (0.0956)	0.0194 (0.0976)	0.0116 (0.0992)	0.0176 (0.0974)	0.0125 (0.0967)	0.0152 (0.0972)
ln unemployment rate	0.3975 (0.8438)	0.3406 (0.8377)	0.3116 (0.8584)	0.2977 (0.8694)	0.3938 (0.8390)	0.4682 (0.8677)
ln population density	7.1828 (10.2482)	8.0755 (10.4734)	8.2447 (10.6499)	7.7068 (10.5954)	7.3667 (10.4002)	7.4176 (10.5427)
ln government debt	0.2255 (1.3540)	0.2285 (1.3617)	0.0815 (1.3859)	0.3221 (1.3671)	0.1446 (1.3679)	0.2693 (1.3634)
corruption control	0.0194 (1.1264)	-0.0005 (1.1336)	-0.1485 (1.1243)	-0.1020 (1.1765)	0.0175 (1.1470)	0.1172 (1.1810)
state-market governance	-0.0631 (0.0371)	-0.0710 (0.0374)	-0.0617 (0.0409)	-0.0744* (0.0376)	-0.0690 (0.0367)	-0.0680 (0.0396)
recession	-0.6550 (0.5436)	-0.6890 (0.5470)	-0.7137 (0.5507)	-0.6723 (0.5393)	-0.6945 (0.5390)	-0.6949 (0.5436)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors clustered at the country level in parentheses. $^*p < 0.05$, $^*p < 0.10$.

5.3 Discussion

5.3.1 Consistency with Theory and Empirical Literature

The empirical association uncovered in this analysis—small in magnitude, positive in sign, and consistently robust across specifications—mirrors the pattern documented in the existing empirical literature. Cross-national contributions (Alonso et al., 2017), longitudinal assessments of local government outsourcing (Petersen et al., 2018), and regional studies (Delgado-Téllez et al., 2022) all converge on the conclusion that outsourcing does not, on average, yield appreciable reductions in public expenditure and may instead be associated with marginal increases. The magnitude of the effect estimated here therefore situates itself squarely within this empirical landscape, reinforcing the view that the fiscal implications of outsourcing, when evaluated at macro-levels of aggregation, tend to be modest and often counter to the efficiency-enhancing expectations that motivated its diffusion.

Interpreting this result requires revisiting the theoretical frameworks reviewed earlier. The mainstream economic and managerial rationale for outsourcing—rooted in the presumed benefits of *ex ante* competition, scale economies, and performance-oriented incentives (Domberger and Jensen, 1997; Brown et al., 2010)—provides the natural baseline against which fiscal effects are typically assessed. As such, it constitutes the reference point for evaluating whether the empirical evidence corroborates the efficiency-centred promise historically associated with contracting out. From this perspective, the positive coefficient reported here does not support the expectation that outsourcing systematically reduces public expenditure.

At the same time, the finding is fully compatible with the internal limits of the mainstream framework itself. Transaction cost economics highlights how the contracting process generates significant administrative, supervisory and coordination demands which may offset, partially or entirely, the potential efficiency gains attributed to private provision, particularly in service domains characterised by relational intensity or outcome uncertainty (Williamson, 1975, 1985; Brown et al., 2010). The empirical pattern identified—modest in scale, stable across model specifications, and not driven by procurement composition—accords with these theoretical predictions and suggests that transaction and monitoring costs remain a salient feature of contemporary outsourcing practices.

Important additional insight is provided by the procurement controls that exhibit statistically significant coefficients. The positive association between subcontracting intensity and expenditure is consistent with the dynamics highlighted in transaction cost theory: multi-layered contractual structures can increase coordination and supervision demands, potentially expanding the number of actors involved and thereby contributing to higher monitoring requirements (Williamson, 1985; Domberger and Jensen, 1997). Likewise, the positive

coefficients on the shares of EU and non-EU bidders may reflect features documented in the empirical literature, as cross-border procurement is often associated with more specialised suppliers, more complex tendering processes, and higher administrative and compliance burdens. While these associations cannot be interpreted causally, they are compatible with theoretical arguments that emphasise how procurement complexity may shape expenditure patterns, even where competitive mechanisms are formally in place.

The only negative coefficient among the procurement controls—the share of service contracts—is small and marginally significant and therefore must be interpreted with caution. However, its sign is not inconsistent with the sectoral evidence. The GPPD “services” category aggregates activities ranging from standardised technical services (e.g. cleaning, waste collection, basic maintenance) to complex relational services (e.g. social care, health-related functions). Existing studies show that potential cost reductions are concentrated almost exclusively in the former group, where tasks are routine-intensive and performance is easily measurable, while outsourcing of relational services generally increases costs due to high monitoring requirements and contractual incompleteness. A slightly negative coefficient is therefore compatible with a composition effect in which standardised services dominate the aggregate category. Crucially, this does not constitute evidence of systematic savings from outsourcing but rather underscores the heterogeneity of service domains and the limitations of aggregate classifications. A more granular CPV-based decomposition would be necessary to distinguish between technical and relational services and to assess sector-specific mechanisms with greater precision.

A broader interpretive framework is provided by the institutional and critical literature examined in the review. These contributions do not directly theorise the fiscal effects of outsourcing, nor are they intended to yield predictions about cost efficiency; rather, they problematise the assumption—central to the mainstream rationale—that outsourcing should be evaluated primarily as an efficiency-enhancing technique. Within this wider body of work, the institutional literature reviewed in Chapter 2 conceives outsourcing as part of a broader restructuring of state capacity rather than as a discrete, efficiency-oriented procurement choice. These accounts do not offer fiscal predictions, but instead question the very premise that efficiency should constitute the main evaluative criterion. As discussed by Collington and Mazzucato (2024), extensive reliance on external providers can progressively dis-embed the state from the organisational and learning processes through which public value is produced, weakening in-house capabilities and generating forms of institutional dependence on contractors. From this perspective, the modest and non-specific association observed in the empirical analysis appears less as a deviation from an expected efficiency benchmark, and more as an indication that the fiscal consequences of outsourcing are embedded within

broader institutional transformations whose effects are diffuse, incremental, and not primarily reducible to variation in aggregate expenditure.

5.3.2 Limitations

Data Structure The empirical analysis is constrained by several features of the dataset. Coverage in the Global Public Procurement Database becomes sufficiently stable only after 2011 (Appendix A), which restricts the usable sample to 2011–2021 and yields a short and unbalanced panel. With a limited time dimension, the fixed-effects estimator relies mainly on year-to-year within-country variation. In such settings, idiosyncratic shocks cannot be averaged out over time, and the variance of $\hat{\beta}_{FE}$ becomes more sensitive to short-run disturbances in both the regressor and the dependent variable. As a result, the estimates reflect high-frequency movements rather than medium-run dynamics.

A second limitation concerns heterogeneity in reporting completeness. The number and value of tenders recorded in the GPPD depend on country-specific administrative and transparency practices. Because the aggregate measure of outsourcing intensity is constructed from reported tenders, measurement error in the regressor cannot be assumed to be purely random. Temporal or cross-country variation in reporting affects the noise component of the aggregate indicator and weakens the stability of the within-country variation on which identification relies.

A further constraint arises from the distribution of procurement values, which is heavily right-skewed and characterised by excess kurtosis (Appendix A.7). Log-transformation reduces, but does not remove, the influence of extreme observations. In heavy-tailed settings, the sample mean is not a robust statistic, and country–year aggregates remain sensitive to unusually large tenders. Consequently, changes in outsourcing intensity may reflect outlier-driven variation rather than systematic shifts in contracting behaviour.

Finally, both outsourcing intensity and several macro-fiscal controls display considerable within-country volatility over the sample period (Table 3.5). High temporal variation reduces the signal-to-noise ratio of the fixed-effects transformation: the systematic component of $X_{it} - \bar{X}_i$ becomes harder to distinguish from transitory noise, and the resulting loss of precision is exacerbated by the short time dimension of the panel.

Taken together, these characteristics imply that the estimated relationship between outsourcing intensity and public expenditure is identified from relatively noisy within-country variation. Although the specification is directional, the structure of the data does not support causal interpretation, and the results should be understood as descriptive associations.

TWFE Identification and Heterogeneity A further limitation concerns the identifying assumptions underlying the two-way fixed-effects (TWFE) specification. The validity of the TWFE specification rests on the assumption that the regressor of interest exerts the same effect for all units and in all time periods. This requirement rules out cross-sectional variation in the underlying relationship, and it relies on a linear additive structure with a single common coefficient. In empirical settings, such uniformity is not commonly observed: units often respond differently to the same covariate. When this occurs, the TWFE estimator does not recover a common parameter but combines heterogeneous effects into a single coefficient, which can lead to biased estimates (Imai and Kim, 2020).

Appendix C.5 provides direct empirical evidence of heterogeneity in this setting. The interacted specification with country-specific coefficients shows substantial dispersion across Member States: several elasticities are positive and significant, others are negative and significant, and many are close to zero. The interquartile range spans approximately -0.006 to 0.018. This pattern contradicts the constant-effect restriction required by the baseline TWFE model. Such dispersion also implies that, even if the pooled TWFE coefficient does not capture a common structural elasticity, outsourcing may exert economically meaningful effects within a non-trivial subset of countries.

Given these limitations, the TWFE estimator cannot be interpreted as a common structural elasticity: under heterogeneous effects, the TWFE coefficient is known to be biased with respect to the parameter it would identify under homogeneity.

Endogeneity Concerns Given the structure of the variables and the empirical setting, an additional limitation concerns potential endogeneity. Indeed, there might be distinct channels through which the outsourcing indicator is correlated with primary expenditure.

First, reverse causality is a substantive concern. Fiscal expansions and consolidations could induce adjustments in outsourcing behaviour: governments under budgetary pressure may rely more heavily on private providers. This simultaneity channel—documented in existing studies—implies that outsourcing intensity may partly respond to the fiscal dynamics the model attempts to explain.

Secondly, omitted time-varying determinants may affect both expenditure and procurement behaviour. Country and year fixed effects absorb time-invariant heterogeneity and common shocks, substantially mitigating the scope for omitted-variable bias. However, they cannot rule out the presence of unobserved factors that evolve over time and influence both outsourcing intensity and expenditure levels. The analysis therefore acknowledges the possibility of residual omitted-variable endogeneity.

Third, simultaneity may also generate endogeneity. In this setting, outsourcing intensity

and primary expenditure can both adjust to the same fiscal conditions within the year. The micro-level construction of the outsourcing indicator attenuates this concern: because it is derived from tender-level procurement data rather than from budgetary aggregates, it is not mechanically linked to the dependent variable. This reduces the likelihood that the regressor incorporates components of public expenditure by construction, although it does not fully remove the possibility of simultaneous adjustment to common fiscal shocks.

In principle, these concerns could be addressed through instrumental-variable strategies or dynamic panel models. In practice, the lack of externally validated instruments in the existing literature and the short time dimension of the panel limit the feasibility of both IV and lagged specifications.

Stationarity Properties A final limitation concerns the time-series properties of the variables. Standard panel unit-root tests—such as Levin–Lin–Chu, Im–Pesaran–Shin, or Fisher-type ADF procedures—require substantially longer and more balanced panels than the eleven-year window available here, and cannot be meaningfully implemented. Consequently, the stationarity of the series cannot be formally established.

Nevertheless, some indirect evidence can be drawn from the data. As shown in Table C.9, the lag-1 autocorrelation of primary expenditure lies between 0.36 and 0.77, with a median around 0.61, indicating a high degree of persistence and limited year-to-year adjustment. While such persistence does not in itself imply non-stationarity, it is consistent with slow-moving fiscal aggregates and suggests that part of the observed co-movement with outsourcing intensity may reflect underlying low-frequency dynamics rather than purely contemporaneous variation.

The use of country and year fixed effects absorbs time-invariant heterogeneity and common shocks, which mitigates the risks associated with potential non-stationarity in the within-country dimension. Given the short time span and the inability to test for unit roots, the estimates should therefore be interpreted with appropriate caution, acknowledging that persistent dynamics may influence the magnitude of the estimated associations.

5.4 Summary of Evidences

The empirical analysis provides a preliminary characterisation of the directional association between outsourcing intensity and primary public expenditure across European countries. The baseline two-way fixed-effects estimates indicate a modest average within-country elasticity, while masking substantial cross-country heterogeneity documented in the extended diagnostics. Robustness checks based on procurement composition controls corroborate the

stability of the aggregate coefficient and offer additional insight into the structure of the underlying data. Decomposition exercises, included for completeness, do not yield interpretable or consistent patterns and therefore play a limited role in the empirical assessment.

Taken together, the results should be interpreted as descriptive evidence of an average association, albeit consistent with macro-aggregate findings in the existing literature. As emphasised in the Discussion, their interpretation requires caution in view of the limitations related to data structure, the heterogeneity and identification constraints of the TWFE design, potential endogeneity concerns, and the stationarity properties of the short panel.

Chapter 6

Conclusion

The purpose of this research project was to examine whether within-country changes in public sector outsourcing are systematically associated with movements in primary public expenditure across EU Member States. The analysis has focused on the fiscal rationale (the expectation that contracting-out should reduce public spending by mobilising private-sector efficiency) since this argument has been recognized as the most recurrent and politically salient justification in orthodox economic literature and managerial debate. For this reason, assessing the empirical validity of this rationale has been considered a necessary benchmark for evaluating the correspondence between policy narratives and observable macro-fiscal patterns. To address this question, the study has employed a two-way fixed-effects framework estimated on a panel of 27 EU Member States over 2011–2021. The outsourcing indicator used is micro-founded, constructed from tender-level procurement data, exploiting a recently released Global Procurement Dataset (GPPD). Due to the chosen dataset, the specification has further incorporated a set of procurement composition controls that reflect structural features of procurement markets.

The results have indicated a small but statistically robust positive elasticity between outsourcing intensity and real primary expenditure. This association has been proven stable across extensive macro-fiscal, institutional and political controls, while it does not appear to operate through identifiable channels in ESA or COFOG aggregates. Consistent with prior cross-national and subnational evidence at the aggregate level, the analysis does not corroborate the expectation that outsourcing systematically reduces public expenditure. Instead, it has identified modest expansionary patterns. The interpretation of this pattern aligns with theoretical approaches that question the assumption of inherent efficiency gains from private provision. Transaction cost and organisational economics emphasise that outsourcing generates coordination, monitoring and contracting burdens that increase with contractual complexity and the relational nature of services. The procurement controls adopted

here reinforce this view: higher subcontracting intensity and greater cross-border supplier presence, both indicative of multi-layered and administratively demanding contracting environments, are positively associated with expenditure. The negative coefficient on service contracts, though weak and statistically marginal, may instead reflect the heterogeneity of this category, where routine-intensive technical services coexist with relational services for which outsourcing is typically cost-increasing.

Clearly, these conclusions must be interpreted in light of several limitations. Indeed, in the analysis the empirical window is short, constrained by the period of stable reporting in the Global Public Procurement Database, and covers years marked by extraordinary fiscal shocks—post-crisis consolidation, austerity cycles, and pandemic-driven expenditure surges (2011-2021). These dynamics may attenuate or obscure relationships that would emerge more clearly in a longer or less dynamic temporal horizon. Additionally, variation in reporting completeness introduces non-random measurement error into the outsourcing indicator, and the heavy-tailed distribution of procurement values renders country-year means sensitive to extreme tenders. The fixed-effects model imposes a homogeneous effect across Member States, despite clear heterogeneity in national coefficients. Endogeneity remains plausible: outsourcing may itself respond to fiscal pressures or unobserved time-varying conditions. Finally, the short time series prevents the estimation of dynamic models capturing lagged expenditure responses to multi-year contracting processes.

These limitations nonetheless point towards promising directions for future research. First of all, the richness of the Global Public Procurement Database (GPPD)—its CPV-based sectoral detail, territorial identifiers and buyer characteristics—suggests the value of multilevel frameworks capable of capturing cross-level mechanisms obscured by country-year aggregation. More granular COFOG analyses could uncover expenditure channels hidden within broad functional aggregates. Indeed, outsourcing may influence specific sub-items within functional categories that aggregate data cannot fully capture. While extending the analysis to additional jurisdictions covered by the GPPD (e.g. USA) and expanding the temporal horizon would support dynamic specifications better suited to modelling the persistence of public expenditure and delayed effects of procurement decisions.

Such extensions would strengthen the analytical scope of this research by deepening the understanding of the fiscal patterns linked to public sector outsourcing and the channels through which they unfold. Nonetheless, it remains essential to acknowledge the inherent limitations of fiscal rationales when interpreting outsourcing dynamics. Explaining and justifying public sector outsourcing as a mere process of cost cut for public expenditure hides a broader conceptualisation of outsourcing not as a narrow cost-containment device, but as a process of state transformation. Indeed, critical institutional analyses emphasise that

extensive reliance on external providers restructures the organisation of public administration, fragments service delivery chains, and relocates operational knowledge and expertise outside public institutions (Weghmann and Sankey, 2022; Collington and Mazzucato, 2024). Over time, these dynamics can erode in-house capabilities, generate forms of organisational dependency, and weaken the state’s capacity for coordination, learning and strategic direction. The labour-market implications are equally significant. In institutional environments with weak collective bargaining coverage or limited regulatory protections, outsourcing has been associated with rising precarity, deteriorating working conditions, fragmented career trajectories and downward pressure on wages—effects that disproportionately affect women and ethnic minorities concentrated in outsourced occupations such as cleaning, catering and security. In countries with stronger collective bargaining systems or legal protections these effects are mitigated but not eliminated: while wage compression may be limited, other dimensions, including overtime, leave entitlements and career mobility, often worsen. Outsourcing therefore contributes to the reconfiguration of employment standards along the public–private boundary, with implications that extend beyond fiscal outcomes.

Taken together, these considerations underscore the need for analytical frameworks that transcend a narrowly fiscal evaluation of outsourcing. The modest and positive elasticity identified in this thesis challenges the empirical validity of the efficiency-based justification historically attached to contracting-out. More fundamentally, the evidence and the broader literature suggest that outsourcing must be understood as a multidimensional institutional process whose long-term consequences concern not only expenditure levels but also the evolution of state capacities, the organisation of public employment, the distribution of expertise and the democratic governance of essential services. While this study has focused on fiscal outcomes, its empirical design does not allow an assessment of three dimensions that are central to the broader implications of outsourcing: the quality and equity of the services provided; the potential crowding-out of internal public investment and administrative capacity; and the political–economy mechanisms, including electoral incentives and corruption risks, that may influence procurement dynamics. These aspects lie beyond the scope of the present analysis but represent important avenues for future research. Future research integrating these dimensions will be essential to understanding outsourcing not merely as a managerial choice, but as a central vector in the contemporary reconfiguration of public administration. In particular, extending the analysis beyond fiscal outcomes to examine how outsourcing affects the quality and security of public employment, collective bargaining arrangements, and the evolution of state capabilities would help situate contracting-out within a broader political–economic assessment of contemporary public administration.

Appendix A

GPPD Diagnostic and Coverage Analysis

A.1 Dataset Overview

The Global Public Procurement Dataset (GPPD) used in this thesis was filtered for the EU-27 countries, covering contracts awarded between 2006 and 2021. After cleaning and harmonisation, the final dataset contains **2,371,162 contracts** across the EU-27. Each observation corresponds to a single awarded contract, including detailed information on buyers, suppliers, contract value, product classification, and procurement procedures.

A.2 Variable Completeness and Quality

To evaluate the robustness of the dataset, a missing-value diagnostic was performed. Core variables such as `buyer_country`, `buyer_nuts`, `tender_year`, `tender_supplytype`, and `tender_digiwhist_price` show complete or near-complete coverage. By contrast, variables related to subcontracting and process duration (e.g. `lot_updateddurationdays`, `bid_subcontractedproportion`) display high missingness (>90%) and were excluded from the analysis.

Table A.1: Top variables by percentage of missing values (EU-27, 2006-2021)

Variable	Missingness (%)
lot_updateddurationdays	99.8
bid_subcontractedproportion	99.2
lot_smebidscount	60.3
corr_singleb	19.4
lot_bidscount	19.4
tender_digiwhist_price	10.3
buyer_country	0.0
buyer_nuts	0.0

A.3 Contract Composition by Type

Contracts are classified into three main supply categories—*Works*, *Supplies*, and *Services*—according to the harmonised CPV structure. Supplies dominate in both number and total value, while Works contracts, though fewer, represent the largest individual projects.

Table A.2: Contracts by category (EU-27, 2006-2021)

Category	Contracts	Total Value (€ billion)	Average Value (€ million)
Supplies	1,339,386	17,977	14.8
Services	662,673	3,969	6.6
Works	360,526	1,459	4.8

A.4 Coverage and Time Span

Not all countries provide full coverage for 2006-2021. Data completeness stabilises from 2011 onwards. Hence, the analysis focuses on the 2011-2021 window to ensure comparability across member states. *Note.* The apparent decline in the number of contracts in 2021 does not represent a real contraction in procurement activity. As documented by the GPPD developers, several member states—most notably Italy—changed their publication systems during 2021, adopting new open-data formats that have not yet been fully ingested into the database. Future updates of the GPPD will retroactively incorporate the missing publications.

Table A.3: Panel comparison before and after restriction

Panel	Years	Contracts	Total Value (€ billion)
Full (2006-2021)	19	2,371,162	23,472.5
Restricted (2011-2021)	11	2,192,453	23,460.1

Restricting to 2011-2021 removes roughly 7% of contracts but only 0.05% of total value, suggesting early years are sparse and low-value.

A.5 Coverage by Country and Year

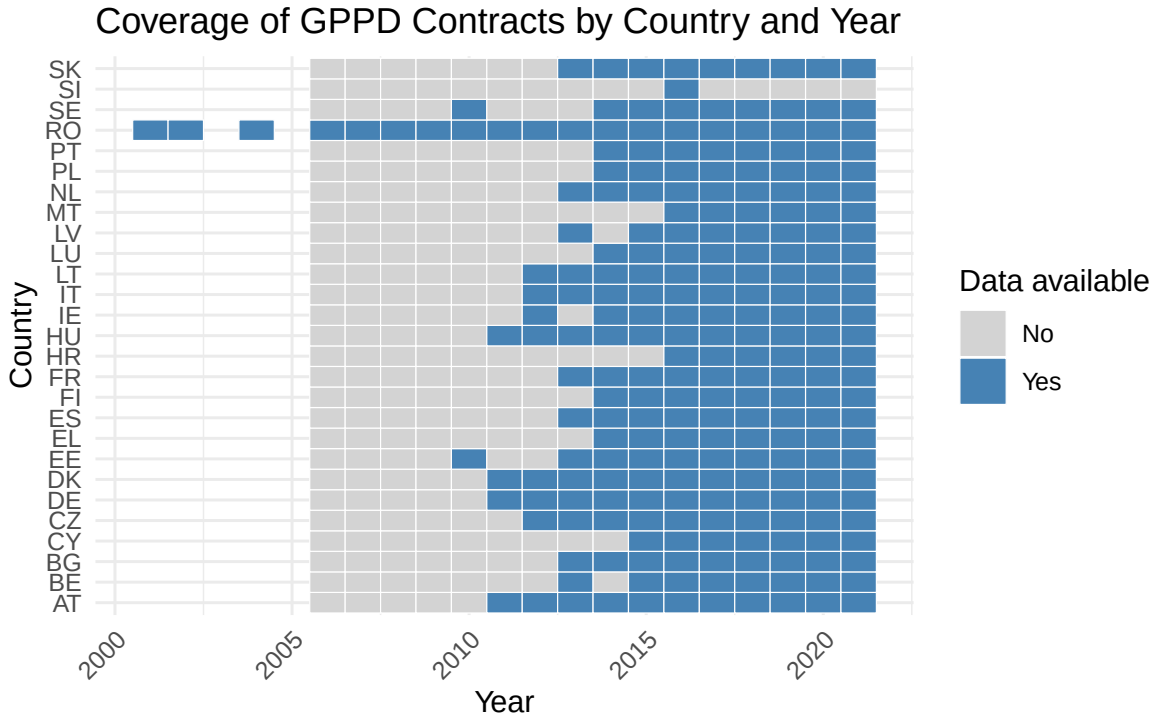


Figure A.1: Coverage of GPPD contracts by country and year (EU-27, 2006-2021). Each blue cell represents a year with available contract data. Coverage becomes stable after 2011 across most member states.

A.6 Identification of the Italy 2016 Outlier

This appendix documents the presence of an extreme outlier in the Italian procurement data for 2016. The anomaly originates from the variable `avg_contract_price`, which spikes to an unusually high level that is inconsistent with the historical series for Italy and with the cross-country distribution.

Using the micro-level prices (`tender_digiwhist_price`), the distribution for Italy in 2016

exhibits a mean of 4.15 billion PPP-adjusted USD and a median of only 1.98 million, with a maximum value above 100 billion. This extreme right tail causes the yearly average to be mechanically inflated.

Table A.4: Summary of contract values for Italy in 2016

Statistic	Value
Mean tender value (<code>tender_digiwhist_price</code>)	4,147,329,894
Median tender value	1,982,376
Maximum tender value	109,200,000,000
Observations	2,748

At the aggregated country–year level, this leads to an observation (`ln_valmean` = 22.1) that lies more than 2.7 standard deviations above the Italian historical mean and does not align with any pattern observed in other member states.

Given its clear mechanical origin and the absence of any economic rationale for such a spike, the observation `Italy-2016` is excluded from the baseline empirical analysis. The exclusion affects only one out of 297 country–year cells and does not alter the time trends or cross-country patterns documented in the main text.

A.7 Distributional Diagnostics: Skewness and Kurtosis of Contract Values

A key characteristic of public procurement microdata is the extreme heterogeneity in contract values. The distribution typically comprises a large mass of small-value contracts and a very small number of high-value tenders, leading to heavy right tails. This feature is well documented in the literature on procurement performance and cost modelling.

To assess the distributional properties of the GPPD contract-value data, skewness and kurtosis were computed for all awarded contracts in the EU-27 between 2006 and 2021. The analysis uses the variable `tender_digiwhist_price`, expressed in PPP-adjusted international dollars, and includes only strictly positive values.

Table A.5: Skewness and kurtosis of contract values (EU-27, 2006–2021), after removing the IT–2016 outlier

Statistic	Raw Values	Log Values
Skewness	597.63	−0.046
Kurtosis	405,713.7	3.884
Excess kurtosis	405,710.7	0.884

After removing the extreme outlier corresponding to Italy in 2016, the raw distribution of contract values remains heavily right-skewed (skewness 598) and extremely leptokurtic (kurtosis 405,714), consistent with the presence of large-value tenders. The logarithmic transformation again yields a distribution with near-zero skewness (−0.046) and moderately positive excess kurtosis (0.884), confirming that log-values are appropriate for aggregation and modelling.

A.8 Summary

The GPPD dataset offers rich, high-quality microdata for cross-country analysis of outsourcing patterns. While some process-level variables exhibit missingness, coverage from 2011 onward is comprehensive, supporting a robust empirical investigation.

Appendix B

Data Sources and Variable Documentation

B.1 Overview

This appendix provides an exhaustive documentation of all variables used in the empirical analysis, specifying their sources, dataset codes, and the main transformations applied. It serves as a technical complement to Chapter 3, which discusses data sources and variables construction. The purpose of this section is to document the construction of indicators and the harmonisation of datasets across EU member states, ensuring transparency and replicability of the analysis.

All series are harmonised at the *country-year* level for the EU-27 and cover the period 2011–2021. Monetary variables are deflated by the GDP deflator (2010 = 100). When relevant, logarithmic and country-demeaned transformations were applied to capture within-country variation over time.

B.2 Dependent Variables

The dependent variables were constructed from Eurostat’s *Government Finance Statistics* (GFS) under the ESA 2010 framework. These indicators combine official expenditure components to isolate specific dimensions of fiscal policy. Nominal series were converted to real terms using the GDP deflator (2010 = 100). For cross-country comparability, all variables were log-transformed and demeaned by country, capturing within-country temporal variation.

Table B.1: Baseline dependent variable

Variable	Source	Code	Transformation
Real primary expenditure (TE – D.41PAY)	Eurostat	gov_10a_main	ln(real value), demeaned by country

Constructed as total general government expenditure (TE) minus interest payments (D.41PAY). Nominal values were deflated using the GDP deflator (base 2010 = 100), and then log-transformed and demeaned to isolate discretionary spending dynamics.

The robustness analysis further decomposes total expenditure into its main economic and functional components, following ESA 2010 and COFOG classifications.

Table B.2: Economic composition of expenditure (robustness)

Variable	Source	Code	Transformation
Intermediate consumption	Eurostat	gov_10a_exp, P2	ln(real value)
Compensation of employees	Eurostat	gov_10a_exp, D1	ln(real value)
Gross fixed capital formation	Eurostat	gov_10a_exp, P5	ln(real value)

Economic aggregates constructed directly from ESA 2010 expenditure categories: (P2) goods and services purchased from the market, (D1) public-sector wages, (P5) investment. All variables deflated and expressed in logs.

Table B.3: Functional composition of expenditure (COFOG)

Variable	Source	Code	Transformation
General public services	Eurostat	TF6_1	ln(real value)
Public order and safety	Eurostat	TF6_3	ln(real value)
Economic affairs	Eurostat	TF6_4	ln(real value)
Health	Eurostat	TF6_7	ln(real value)
Education	Eurostat	TF6_9	ln(real value)
Social protection	Eurostat	TF6_10	ln(real value)

COFOG Level 1 aggregates capturing expenditure by functional purpose. Nominal values were deflated using the GDP deflator and converted to natural logs.

B.3 Independent Variable

The key explanatory variable, *outsourcing intensity*, was derived from the Global Public Procurement Database (GPPD) developed by Fazekas & Tóth (2021). Contract-level microdata

were filtered to include only finalised, awarded tenders and winning bids (`tender_isawarded = TRUE`, `bid_iswinning = TRUE`). Contract values (`tender_digiwhist_price`) are expressed in PPP-adjusted international dollars, allowing direct cross-country comparison. The indicator was computed as the mean contract value per country-year and log-demeaned to capture deviations from each country's long-run average.

Table B.4: Independent variable

Variable	Source	Fields	Transformation
Outsourcing intensity	GPPD (Fazekas & Tóth 2021)	<code>tender_digiwhist_price</code> , <code>tender_isawarded</code> , <code>bid_iswinning</code>	ln(PPP-adjusted mean value), demeaned by country

Computed from awarded and winning tenders; PPP-adjusted to ensure cross-country comparability.

B.4 Procurement Composition Controls

To account for structural heterogeneity in procurement markets, a set of control variables was constructed from GPPD microdata. Each variable represents the share of contracts meeting specific characteristics (e.g., services, quality-based awards, subcontracting, foreign suppliers). These indicators summarise compositional shifts in procurement behaviour and are computed as proportions in the interval $[0, 1]$.

Table B.5: Procurement composition variables (GPPD)

Variable	Construction (GPPD fields)	Description
<code>prop_services</code>	Share of <code>tender_supplytype = "services"</code>	Service contracts
<code>prop_works</code>	Share of <code>tender_supplytype = "works"</code>	Works / construction contracts
<code>prop_meat</code>	Share of <code>tender_selectionmethod = "MEAT"</code>	Quality-based awards
<code>prop_subcontracted</code>	Share of <code>bid_issubcontracted = TRUE</code>	Subcontracted tenders
<code>prop_eu_bidders</code>	Share of foreign EU suppliers (<code>bidder_country</code> <code>buyer_country</code> EU)	EU cross-border suppliers
<code>prop_non_eu_bidders</code>	Share of foreign non-EU suppliers	Non-EU suppliers

Computed as proportions over the total number of awarded and winning tenders by country-year. These variables are not demeaned or transformed, as they represent bounded shares capturing the structural composition of procurement.

B.5 Control Variables

Control variables were constructed from harmonised macroeconomic, fiscal, and institutional datasets. All variables were aligned to the GPPD time coverage (2011–2021). Bounded indices and dummies enter in levels; all other continuous variables are expressed in logarithmic form. Annual data were drawn primarily from Eurostat and complemented by the World Bank and the Manifesto Project Database.

Table B.6: Macroeconomic, demographic, and fiscal controls

Category	Variable	Source	Code	Transformation
Macroeconomic	Recession dummy	Eurostat	Derived from <code>nama_10_gdp</code> (B1GQ)	1 if real GDP growth < 0
Labour market	Unemployment rate	Eurostat	<code>une_rt_a</code>	ln(% labour force)
Demographics	Population density	Eurostat	<code>tps00003</code>	ln(inh./km ²)
Fiscal stance	Government debt	Eurostat	<code>gov_10dd_edpt1</code>	ln(% of GDP)

Series retrieved from Eurostat annual datasets under ESA 2010. Nominal ratios were converted to logarithmic form to ensure comparability across countries.

Table B.7: Institutional and political controls

Category	Variable	Source	Field	Transformation
Institutional quality	Control of corruption	World Bank – WGI	<code>CC. EST</code>	Raw index (−2.5 → +2.5)
Political orientation	State–market orientation	Manifesto Project Database	<code>planeco</code> , <code>markeco</code>	Continuous seat-weighted index

Institutional quality measured by the WGI *Control of Corruption* index. Political orientation constructed as the seat-weighted mean of government parties’ positions on economic planning (`planeco`) and market regulation (`markeco`).

B.6 Summary

All indicators were systematically constructed from harmonised, publicly available data sources: Eurostat for fiscal, macroeconomic, and demographic statistics; the Global Public Procurement Database for outsourcing and procurement composition; the World Bank’s Worldwide Governance Indicators for institutional quality; and the Manifesto Project Database for political orientation. Each variable was generated through reproducible transformations, ensuring full transparency and cross-country comparability of the empirical dataset.

Appendix C

Diagnostic Tests for Model Specification

This appendix reports the diagnostic tests supporting the econometric specification adopted in Chapter 4. The analysis includes model-selection tests, heteroskedasticity and serial-correlation diagnostics, cross-sectional dependence, and persistence measures for the dependent variable.

C.1 Model Selection Tests

Hausman Test (FE vs. RE)

The Hausman test compares fixed-effects and random-effects estimates under the null hypothesis that regressors are uncorrelated with country-specific unobservables. Rejection implies that random effects are inconsistent and supports the fixed-effects estimator.

Table C.1: Hausman Test: FE vs. RE

	χ^2	df	p-value
Hausman statistic	49.653	7	1.69×10^{-8}

F-tests for Country and Time Fixed Effects

These tests evaluate whether country and time fixed effects are jointly significant. Rejection of the null indicates that pooled OLS is misspecified relative to a fixed-effects model.

Table C.2: F-test for Country Fixed Effects

	Statistic	df ₁	df ₂	p-value
F-test	14.388	26	197	$< 2.2 \times 10^{-16}$

Table C.3: F-test for Time Fixed Effects

	Statistic	df ₁	df ₂	p-value
F-test	52.418	10	213	$< 2.2 \times 10^{-16}$

Breusch–Pagan LM Test for Random Effects

The Breusch–Pagan LM test examines whether the random-effects variance component is statistically different from zero. The null is not rejected, consistent with a negligible variance.

Table C.4: Breusch–Pagan LM Test for Random Effects

Statistic	df	p-value
0.034	1	0.8527

C.2 Error Structure Diagnostics

Heteroskedasticity

The studentised Breusch–Pagan test evaluates the null hypothesis of homoskedastic idiosyncratic errors in the fixed-effects residuals.

Table C.5: Breusch–Pagan Test for Heteroskedasticity

BP statistic	df	p-value
19.841	7	0.00593

Serial Correlation

The Wooldridge / Breusch–Godfrey test examines first-order serial correlation in panel residuals. Rejection of the null indicates AR(1) autocorrelation.

Table C.6: Wooldridge Test for Serial Correlation

χ^2	df	p-value
44.597	1	2.42×10^{-11}

Cross-Sectional Dependence

Pesaran’s CD test evaluates whether residuals are correlated across countries, which may occur due to common shocks or unobserved global factors.

Table C.7: Pesaran CD Test for Cross-Sectional Dependence

z-value	p-value
-2.173	0.0298

C.3 Persistence Diagnostics

Country-level lag-1 autocorrelation coefficients are computed for the demeaned dependent variable, documenting its time-series persistence over 2011–2021. Coefficients fall between 0.35 and 0.77 for almost all countries, indicating moderate inertia in primary expenditure.

Table C.8: Lag-1 Autocorrelation of Demeaned Primary Expenditure

Country	AC(1)	Observations
AT	0.645	11
BE	0.638	8
BG	0.477	9
CY	0.561	7
CZ	0.723	10
DE	0.705	11
DK	0.557	11
EE	0.688	9
EL	0.357	8
ES	0.650	9
FI	0.613	8
FR	0.580	9
HR	0.585	6
HU	0.738	11
IE	0.641	9
IT	0.524	9
LT	0.699	10
LU	0.657	8
LV	0.517	8
MT	0.530	6
NL	0.612	9
PL	0.682	8
PT	0.522	8
RO	0.769	11
SE	0.564	8
SI	—	1
SK	0.526	9

Table C.9: Distribution of Lag-1 Autocorrelation Coefficients

Min	1st Qu.	Median	Mean	3rd Qu.	Max	NAs
0.3574	0.5364	0.6121	0.6062	0.6757	0.7694	1

C.4 Additional Dynamic Specification Checks

This section reports the diagnostic tests associated with the dynamic specification discussed in Chapter 4. The aim is to evaluate whether the inclusion of a lagged dependent variable

raises consistency concerns for the fixed-effects estimator in the present data setting.

Hausman Test: FE vs. FE-IV (Anderson-Hsiao)

The FE estimator including the lagged dependent variable is compared with an Anderson-Hsiao FE-IV specification instrumenting $\text{lag}(y_{it}, 2)$. The test does not reject the null of coefficient equality, providing no statistical evidence of inconsistency of the FE estimator.

Table C.10: Hausman Test: FE vs. FE-IV (Anderson-Hsiao)

χ^2	df	p-value
1.6425	2	0.4399

Variance Inflation Factors (Dynamic FE)

Variance inflation factors from the dynamic FE specification indicate no multicollinearity concerns. All VIF values are well below conventional thresholds.

Table C.11: Variance Inflation Factors: Dynamic FE Model

Variable	VIF
Lagged dependent variable	1.476
X_{it}	1.254
Log unemployment rate	2.121
Log population density	1.351
Log government debt	1.620
Control of corruption	1.080
State-market orientation	1.135

C.5 TWFE Heterogeneity: Country-Specific Effects

This section reports diagnostic evidence on cross-country heterogeneity in the elasticity of real public expenditure with respect to outsourcing intensity. Table C.12 document substantial dispersion in the country-specific coefficients obtained from an interacted model of the form:

$$\ln Y_{it} = \alpha_i + \tau_t + \sum_c \beta_c (X_{it} \times \mathbb{1}\{i = c\}) + \gamma' Z_{it} + \varepsilon_{it}.$$

The coefficients β_c represent deviations from the reference country (Italy), for which the effect is normalised to zero. Standard errors are clustered at the country level.

Country-Specific Coefficients

Table C.12: Country-Specific Elasticity of Public Expenditure w.r.t. Outsourcing Intensity

Country	Estimate	Std. Error	p-value	CI Low	CI High
AT	0.0008	0.0066	0.904	-0.0127	0.0143
BE	-0.0061	0.0060	0.315	-0.0185	0.0062
BG	0.0191	0.0114	0.105	-0.0043	0.0424
CY	0.0824	0.0378	0.0385	0.0047	0.1600
CZ	-0.0058	0.0043	0.192	-0.0146	0.0031
DE	0.0045	0.0086	0.601	-0.0131	0.0222
DK	0.0099	0.0075	0.193	-0.0054	0.0253
EE	0.0406	0.0067	0.0000	0.0268	0.0544
EL	-0.0167	0.0045	0.0010	-0.0259	-0.0074
ES	0.0076	0.0062	0.232	-0.0052	0.0203
FI	0.0119	0.0068	0.0908	-0.0020	0.0259
FR	-0.0054	0.0073	0.467	-0.0204	0.0096
HR	-0.0143	0.0194	0.466	-0.0542	0.0255
HU	0.0143	0.0144	0.331	-0.0154	0.0440
IE	0.0545	0.0178	0.0051	0.0178	0.0911
IT (ref.)	0	—	—	—	—
LT	0.0598	0.0085	$< 10^{-6}$	0.0423	0.0773
LU	-0.0156	0.0069	0.0324	-0.0297	-0.0014
LV	0.0178	0.0057	0.0043	0.0061	0.0295
MT	0.0459	0.0388	0.247	-0.0338	0.1260
NL	-0.0014	0.0042	0.739	-0.0101	0.0072
PL	0.1570	0.0292	0.0000	0.0972	0.2170
PT	-0.0238	0.0074	0.0037	-0.0391	-0.0085
RO	0.0066	0.0095	0.495	-0.0130	0.0262
SE	-0.0115	0.0045	0.0169	-0.0207	-0.0022
SK	0.0018	0.0120	0.884	-0.0229	0.0264

Distributional Summary

The distribution of the estimated country-level elasticities is summarised in Table C.13. The coefficients range from -0.024 (Portugal) to 0.157 (Poland), with a median close to zero and a positive mean.

Table C.13: Distribution of Country-Specific Effects

Min	Q1	Median	Mean	Q3	Max
-0.0238	-0.0056	0.0046	0.0161	0.0184	0.1570

Classification by Sign and Significance

Table C.14: Classification of Country Effects

Group	Countries
Positive-Significant	6
Negative-Significant	4
Positive-Not Significant	10
Negative-Not Significant	5
Reference (IT)	1

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